



GE Medical Systems

Technical Publications

Direction 2248332

Revision 1

Magnet and System Deinstallation for Upgrade to Signa^R WideOpen (HiSpeed & EchoSpeed)

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785-5052/8*323-5052 **immediately** after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section "S" of the Policy & Procedure Bulletins.

3/12/92



GE Medical Systems

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
Rev A . . .	Aug 6, 1999 . . .	Preliminary Revision
Rev B . .	Aug 31, 1999 . . .	Section 1 – Material Disposition Guidance, added "Type of magnet being returned". Section 3 – Revised flowcharts to include return of EDM Magnet Monitoring equipment, retaining of Patient Table information and other minor changes. Section 4 – Revised cable removal table, added re-labeling and re-routing tables, and revised cable removal maps. Section 5 – Added information of removing cryogen meters for 8.x System cabinets.
Rev C . .	Sep 17, 1999 . . .	Section 3 – Added note on retaining head coil, surface coils, and pads for reuse on new system. Section 4 – Added magnet cables to removal chart. Updated other cable removals. Updated cable removal maps accordingly. Section 5 – Added note on saving head coil, surface coils and pads for reuse on upgraded system. Added step on saving RF extension cables, if required.
Rev 0 . .	Oct 14, 1999 . . .	Incorporated information obtained from first field upgrade. Section 1 – Added new Equipment Return Process. Added items to the return checklist. Section 3 – Added notes for retaining preamps, head coil, surface coils and pads. Added note for Equipment Return Process. Inquiry about customer options and parts of Magnet room to be deinstalled. Section 4 – Revised cable removal table and cable removal maps. Section 5 – Added instructions for returning of body coil in magnet. Revised penetration panel upgrade procedure.
Rev 1 . . .	Dec 9, 1999 . . .	Added Direction 2147446, <i>Signa Horizon 1.5T & 1.0T UFI-Compatible Surface Coils Instructions</i> to front of manual. Revised page 3–5 by adding Notes 11 and 12 (SPR#:MRlge57187).

LIST OF EFFECTIVE PAGES

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Product Description:

The following procedures need to be followed for all new and upgrade 1.5T Signa Horizon system installations.

Note

The Surface Coil UFI-compatibility issue and warning labels plan do **NOT** apply to 1.0T Signa Horizon systems.

With the purchase or upgrade of a 1.5T Horizon, HiSpeed, or EchoSpeed, there are several RF coils that will no longer be compatible with the Signa Horizon system. It is required that these non-compatible coils be removed from the site and **returned** to GE. A new coil, of the same type, will be supplied for each of these non-compatible coils before they are removed from the site. This will be accomplished by a marketing promotion, special no-cost S-CAT's, or FMI.

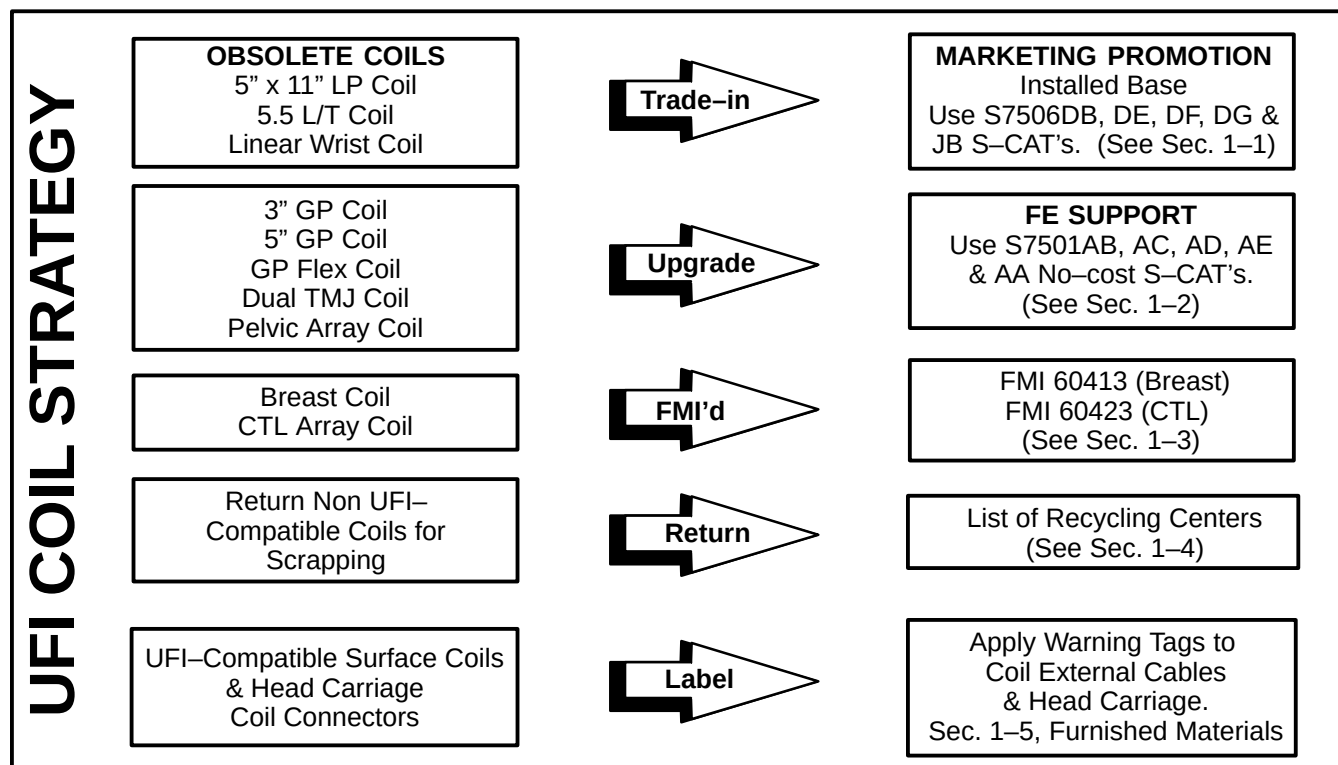
Note

These procedures **do not** apply to 1.5T Signa "Advantage" 5.5 Software Upgrade only systems. For these systems, selecting the "Advantage" Gradient Configuration file option during Load from Cold, will set B1 RMS to the appropriate value.

For NON-GE Approved surface coils: Refer the customer to the original coil manufacturer for information on coil compatibility with Signa Horizon.

Note

GE makes no claims as to compatibility of coils not approved by GE for use on Signa systems. We have no way of knowing if any of these non-GE coils are compatible with 5.5 software and/or hardware. The customer uses these coils at their own risk, and GE accepts no responsibility for their use.

Service Readiness:

4-1 MARKETING PROMOTION

The Following Coils Are Not UFI-compatible. These coils should **NOT** have any UFI-compatible warning tags attached: (See Table.)

The 5" x 11" LP, 5.5" L/T, and Linear Wrist Coils will be replaced through an Installed Base Marketing Promotion to the customer. The customer will receive a letter that will allow him to trade-in the LP and/or L/T coils for a GP Flex coil or Quad T/L coil. The customer will be able to trade-in the Linear Wrist coil for a Quad Wrist coil (E-CAT). This promotion must occur through the GE Sales force. Trade-in allowances and prices are in effect for orders placed through December 31, 1997 and installed by March 31, 1998. This promotion expires on December 31, 1997.

Each of these Trade-in Allowances requires the physical return of a customer owned surface coil(s). Depending on whether one or two coils are being traded-in, the GE field engineer will remove the existing non-compatible coil(s) at the time of the new coil installation.

UFI Non-Compatible Coils:

The non-compatible coils should be returned to your local recycling center for scrapping (See Section 1-4).

COIL (1.5T)	OLD M-CAT# (DELISTED)	OLD GE #
1.5T 5" X 11" LP (Obsolete)	M1085AF	46-258882G1 46-306773G1
1.5T 5.5" Planar L/T (Obsolete)	M1085AJ	46-265503G1
1.5T Linear Wrist Coil (Obsolete)	M1085BA	46-317224P1 46-328444P1

The FRU strategy for the LP and L/T coils left in the field after the promotion is to replace these coils with a new GP Flex coil. The FRU strategy for the Linear Wrist coils left in the field after the promotion is to replace this coil with a similar Linear Wrist coil. The FRU strategy for the Quad Wrist coils in the field is to replace this coil with a similar Quad Wrist coil.

UFI Compatible Coils:

UFI COMPATIBLE TRADE-IN OPTIONS
With one coil, LP or L/T: Upgrade to GP Flex; Order S7506DB Upgrade to Quad T/L; Order S7506DF
With both LP and L/T: Upgrade to GP Flex; Order S7506DE Upgrade to Quad T/L; Order S7506DG
Upgrade to Quad Wrist; Order S7506JB

4-2 FIELD UPGRADES

The Following Coils Are Not UFI-compatible. These coils should **NOT** have any UFI-compatible warning tags attached: (See Table on next page.)

The 3" GP, 5" GP, GP Flex, Dual TMJ Array (with integrated 3" GP coils), and Pelvic Phased Array coils can be replaced with new UFI-compatible coils (S-CAT's). All of these affected coils for each upgraded 1.5T Signa Horizon system needs to be identified before or at the beginning of the system installation. Coils shared between other systems at the site are covered by this process, but there is a limit of one (1) of any Pelvic or Dual array coil and two (2) of any 3" GP, 5" GP, or GP Flex used at the site unless further approval is obtained. If the customer wishes to upgrade additional site coils to UFI-compatible coils, the new UFI-compatible coil M-CAT must be ordered through the site sales representative.

For US Customers only: When these coils are identified, contact Bob Healy at GE in Milwaukee and let him know which coils they are (S-CAT's) and how many of each. Bob will add replacement coils onto the customer's FDO for appropriate delivery to the site. **Bob can be contacted by FAX at 414-827-3348, E-Mail HEALY-R.** Bob can also be contacted by phone with any questions at (414) 827-3284.

For Non US Customers: When these coils are identified, contact your site sales representative to verify that these coils (S-CAT's) have been added to the FDO.

UFI Non-Compatible Coils:

The non-compatible coils should be returned to your local recycling center for scrapping (See Section 1-4).

COIL (1.5T)	OLD M-CAT# (DELISTED)	OLD GE #
1.5T 3" Round GP	M1085AG	46-265311G1 46-307144G1
1.5T 5" Round GP	M1085AH	46-258773G1 46-307080G1
1.5T GP Flex Coil	M1085GP	46-317492G1
1.5T Dual TMJ	M1087AD	46-317395G1
1.5T Pelvic Array Coil	M1087AC (FW50 '95)	46-317000G1 46-320300G1 46-328072G1

Note

On M1087AC with M1087AT, the 1.5T ATD-III, remove the interface board, 46-328131P1 or 46-328131P2, from the Pelvic Array for use in your replacement M1087PA. The M1087PA does not contain an interface board.

UFI Compatible Coils, S-CATS:

COIL (1.5T)	NEW M-CAT#	ORDER S-CAT# (UFI GE #)
1.5T 3" Round GP	M1085GA	S7501AB (2127315)
1.5T 5" Round GP	M1085HA	S7501AC (2127316)
1.5T GP Flex Coil	M1085GR	S7501AD (2128554)
1.5T Dual TMJ	M1087DA (Adapter Only)	S7501AE (2127315 (2)) (Adapter + 3" GP coils (2))
1.5T Pelvic Array Coil	M1087PA	S7501AA (2144100)

4-3 FMI's

The Following Coils Are **Not** UFI-compatible. These coils should **NOT** have any UFI-compatible warning tags attached: (See Table.)

The Breast Coil is being upgraded by a Mandatory FMI. The CTL coil is being upgraded by a Mandatory Safety FMI. Please verify that your site has had both FMI's completed. If not, please contact your local FMI coordinator to request the FMI kits.

UFI Non-Compatible Coils:

The non-compatible coils should be returned in the FMI shipping carton to GPO for upgrading and return to FMI pool as soon as possible.

COIL (1.5T)	OLD M-CAT# (DELISTED)	OLD GE #
1.5T Breast Coil	—	46-320785P1
1.5T CTL Coil	M1087AB	46-301800G1 46-320971G1

UFI Compatible Coils:

COIL (1.5T)	NEW M-CAT#	NEW UFI GE#	COIL STATUS
1.5T Breast Coil	M1085BR	46-320785P2	Coil is upgraded by FMI#60413. (In Process) (Mandatory FMI)
1.5T CTL Coil	M1087CT	2132617 2104799	Coil is upgraded by FMI#60423. (Available FW44 '95) (Safety FMI)

4-4 RETURN NON UFI-COMPATIBLE COILS FOR SCRAPPING:**For GEMS-AM:**

Return surface coil(s) to: GE Recycle Center
2200 E. College Ave.
Cudahy, WI. 53110
Att: Paul Neumiller (414) 747-6997

For GEMS-E:

Return surface coil(s) to: Site Recyclage GEMS-E
300 Rue Fourny
78530 Buc – France
Att: Annie Thomas, Phone # 30704742

For GEMS-A:

Return surface coil(s) to: Your country's local GEMS Recycling Center or one of the above recycling centers.

4-5 FURNISHED MATERIALS

A kit, 2146243 (FRU-2), is attached to inside front frame of the RF/PEN Cabinet above the RF System Controller Module. This kit includes:

Item	Part Number	FRU	Name	Quantity	Description (Remarks)
1	2145936	X	INSTALLATION	1	INSTALLATION DIRECTIONS
2	2146833	2	VINYL STRIPS	20	WHITE VINYL STRIP, 1" X 5", ADHESIVE BACK
3	2146832	2	WARNING TAG	20	UFI WARNING TAG, ADHESIVE BACK
4	2146831	2	WARNING LABEL	2	UFI WARNING LABEL, ADHESIVE BACK, ENGLISH
5	2146831-2	2	WARNING LABEL	2	UFI WARNING LABEL, ADHESIVE BACK, SPANISH
6	2146831-3	2	WARNING LABEL	2	UFI WARNING LABEL, ADHESIVE BACK, PORTUGUESE
7	2146831-4	2	WARNING LABEL	2	UFI WARNING LABEL, ADHESIVE BACK, ITALIAN
8	2146831-5	2	WARNING LABEL	2	UFI WARNING LABEL, ADHESIVE BACK, GERMAN
9	2146831-6	2	WARNING LABEL	2	UFI WARNING LABEL, ADHESIVE BACK, FRENCH
10	2146831-7	2	WARNING LABEL	2	UFI WARNING LABEL, ADHESIVE BACK, CHINESE
11	2146831-8	2	WARNING LABEL	2	UFI WARNING LABEL, ADHESIVE BACK, JAPANESE

SECTION 1 – GETTING STARTED

1-1 SIGNA WideOpen SYSTEM CONFIGURATION

This manual provides instructions to deinstall an existing system in preparation for upgrading to a new Signa® WideOpen (MR/i or CV/i) System. This deinstallation manual is applicable to field strengths of 1.5T and 1.0T.



FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-2 EQUIPMENT RETURN PROCESS

Getting the de-installed equipment sent back to the correct location is another essential step in the Magnet Upgrade process. The de-installed equipment may be sent back to a variety of places depending upon the type of upgrade being performed. Advance planning of equipment return will aid in the upgrade process. The three categories of equipment to be returned are as follows:

1. P & E hardware - i.e. InSite hardware, EDM Magnet Monitor Hardware.
2. System cables, cabinets, magnet enclosure, and miscellaneous hardware.
3. Magnet and associated magnet subsystem components.

For Magnet Monitor return (P&E Hardware) Refer to Direction 2108906, Magnet Monitoring Subsystem (Restricted), SECTION 7, DE-INSTALLATION PROCEDURES, for process and contacts for returning the equipment.

For all other equipment returns use the following process:

1. Contact the MR Product Specialist at GoldSeal Logistics phone number 414-768-7960.
2. Provide the MR Product Specialist with the Magnet type and system hardware configuration being de-installed. If GoldSeal is accepting the returned equipment then all transportation arrangements, empty equipment dollies, coil carts, skids, and packaging will be processed through them. If GoldSeal is not accepting the return then they will advise you to contact GEMS Renewable Resources at phone number 414-747-6997. In this case, GEMS Renewable Resources will be responsible for coordinating the transportation arrangements, empty equipment dollies, coil carts, skids, and packaging.

Note

All equipment being returned to GoldSeal will be sold as used equipment. Special care must be taken by the mechanical de-installers not to cut any cables or damage any of the hardware to be returned.

The equipment will generally be returned on different trucks. The following lists may be useful in the collection and coordination of the equipment to be returned.

1-2 EQUIPMENT RETURN PROCESS (continued)**Truck A - P & E Hardware**

- ☐ EDM and mounting hardware
- ☐ InSite Modem and external Power Supply
- ☐ Magnet Monitor Modem and external Power Supply
- ☐ Modem Cables
- ☐ Magnet Monitor Sensor Kit - Water Flow/Temperature Sensor, Pressure Transducer, Reed Switch and Key Switch
- ☐ Magnet Monitor Cables (equipment room and magnet room)
- ☐ Magnet Monitor Interface Box, external AC/DC Power Supply and ribbon cables

Truck B - System Hardware**Note**

The list will vary depending on the type of electronics upgrade performed.

- ☐ System Cabinet
- ☐ RF Cabinet
- ☐ Gradient Cabinet
- ☐ Penetration Cabinet
- ☐ Penetration Panel
- ☐ Lytron - Heat Exchanger (air to water)
- ☐ Heat Exchanger Fluids (empty container and ethylene glycol label provided with upgrade hardware)
- ☐ Operator Console
- ☐ System Computer
- ☐ Magnet Enclosure Frame & mounting hardware
- ☐ Magnet Enclosure - All fiberglass parts and trim skirts
- ☐ Front & Rear Endbells
- ☐ Bridge & Carriage Cover
- ☐ Rear Pedestal & Electronic Components mounted within
- ☐ Rear Pedestal Covers
- ☐ Dock Assembly
- ☐ PAC Module
- ☐ SRI Module
- ☐ Patient Comfort Module
- ☐ RF/Gradient Body Coil
- ☐ System Cables removed from the Equipment Room
- ☐ System Cables removed from the Magnet Room
- ☐ System Cables removed from Operator Room
- ☐ Phantoms
- ☐ Non-conforming surface coils and extremity coils.
- ☐ Patient Table (optional - only if customer ordered new one and does not want the old one)

1-2 EQUIPMENT RETURN PROCESS (continued)**Truck C - Magnet & Subsystem Components****Note**

All Oxford and S-I magnets will be returned per instructions supplied by GEMS Renewable Resources for disposal.

- ☐ Magnet with cryogenics
- ☐ Cryogen Meters, instrumentation cables and power cable(s).
- ☐ ERU/MRU, instrumentation cable and power cable
- ☐ Magnet vent pipe and fittings
- ☐ Magnet Ramp Supply, Ramp Cables and Ramp Lead Extensions or Current Probes (Used on Oxford & S-I Magnets)
- ☐ Magnet Shim (Resistive) Power Supply and Shim Cables (Used on Oxford and optional for S-I, S-II, S-III and S-IV Magnets)
- ☐ Magnet Shim (Superconducting) Power Supply and Shim Cables (Used on S-I Magnets)
- ☐ Magnet Shim (Superconducting) Power Supply and Shim Cables (Used on S-I Magnets)
- ☐ Shield Cooler Compressor, Compressor Power Cable, Coldhead Power Cable, Supply & Return Gas Lines, and Remote Control Box (Balzers only).
- ☐ Magnet spares kit consisting of burst disc, baffles, vacuum release tool, o-rings, miscellaneous fittings etc.
- ☐ Magnet service ladder
- ☐ Magnet service platform
- ☐ Resistive Shim Filter Box on Penetration Panel
- ☐ Cable support bracket (S1 Magnet)
- ☐ Field mapping fixture

When calling, the following information is needed:

- FDO (Future Delivery Order) number under which the upgrade was shipped
- Site name, address, and phone number
- Type of magnet being returned
- Date of completed Horizon upgrade.
- Name and Beeper Number of Field Engineer that could be available when the equipment is picked up.

1. Prepare the return items for shipment as follows:
 - a. Package other removed modules and boards in a cardboard box and secure inside System Cabinet (or any other cabinet where they fit).
 - b. Use ramp to move cabinets on to shipping pallets. Three installers are required for this heavy lifting task. Fasten with shipping brackets that were removed previously from new cabinets.
2. Process product locator cards for all serialized components removed from the system according to policy.
3. Regulations governing the disposal of this material being deinstalled vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all deinstalled materials per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Renewable Resources for assistance.

1-3 RETURN ITEMS CHECKLIST – BY ROOM

The following items are designated to be returned after the upgrade. All materials should be able to be packaged for return on the two provided cover dollies and pallets. Refer to Section 1-2 , EQUIPMENT RETURN PROCESS for truck assignments.

Magnet Room

- ☐ 1. Magnet
- ☐ 2. Magnet Enclosure Front and Rear Cover
- ☐ 3. Magnet Enclosure Side Covers (4)
- ☐ 4. Magnet Enclosure Frame
- ☐ 5. Front End Bell
- ☐ 6. Rear End Bell
- ☐ 7. Left and Right Side Slide Support assemblies
- ☐ 8. Service Platform
- ☐ 9. Rear Pedestal Frame
- ☐ 10. Rear Pedestal Covers
- ☐ 11. Gradient/RF Coil in shipping container
- ☐ 12. Bore Lamp Asm with Fiber Optic bundle
- ☐ 13. Fiberglass Bridge
- ☐ 14. Obsolete Cables
- ☐ 15. Heat Exchanger (later systems only)
- ☐ 16. Coolant removed from Heat Exchanger and Body Coil (later systems only)
- ☐ 17. EDM Magnet Monitor Sensor Cables
- ☐ 18. ERU/MRU Module

Depending on the type of system being upgraded, some of the items listed below may not be returned. Refer to the appropriate Upgrade Flowchart in Section 3 for details.

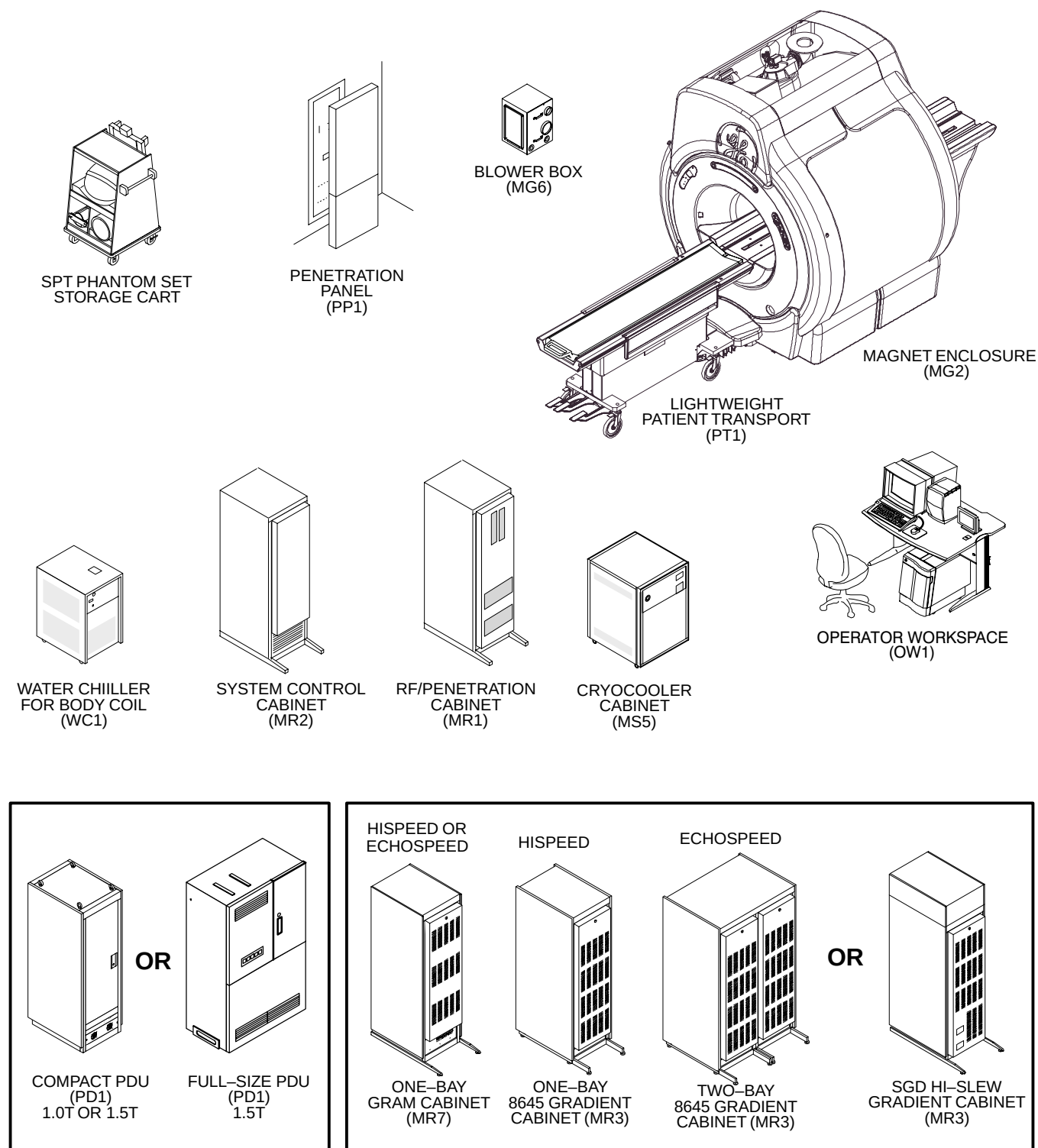
Equipment Room

- ☐ 1. Obsolete Cables
- ☐ 2. Penetration Panel
- ☐ 3. System Cabinet
- ☐ 4. Gradient Cabinet
- ☐ 5. RF Cabinet
- ☐ 6. Penetration Cabinet
- ☐ 7. Computer Cabinet
- ☐ 8. Balzer/Leybold Shield Cooler Compressor, Gase Lines, and Cold Head Power Cables
- ☐ 9. EDM Magnet Monitor Computer, Water Flow Switch, and Cables

Operator Room

- ☐ 1. Console
- ☐ 2. Obsolete Cables
- ☐ 3. Balzer Remote Box

1-4 SYSTEM CONFIGURATION AFTER UPGRADE



SIGNA WideOpen SYSTEM AFTER UPGRADE

ILLUSTRATION 1-1

SECTION 2 – POWER OFF PROCEDURE

2-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Verify that System performance baseline measurements have been obtained. and center frequency should be recorded before system is shut down.
2. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
3. Press the FULL OFF button on the PDU front control panel.
4. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
5. Locate and shut off main disconnect supplying power to the PDU.
6. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
7. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 3 – SYSTEM UPGRADE DEINSTALLATION PROCESS

3-1 DEINSTALLATION FLOWCHARTS

This section contains information on:

<u>TITLE</u>	<u>PAGE</u>
SIGNA WideOpen 8.x SYSTEM DEINSTALLATION FLOWCHARTS	
Signa 3.x to Signa WideOpen 8.x Upgrade Process	3-3
Signa Advantage 4.x to Signa WideOpen 8.x Upgrade Process	3-3
Signa Advantage 1.5T & 1.0T 5.x to Signa WideOpen 8.x Upgrade Process	3-3
Signa Horizon 5.x to Signa WideOpen 8.x Upgrade Process	3-5
Signa Horizon 8.x to Signa WideOpen 8.x Upgrade Process	3-7



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT, CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS
REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE
STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.
KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

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SIGNA WideOpen 8.x UPGRADE FROM:

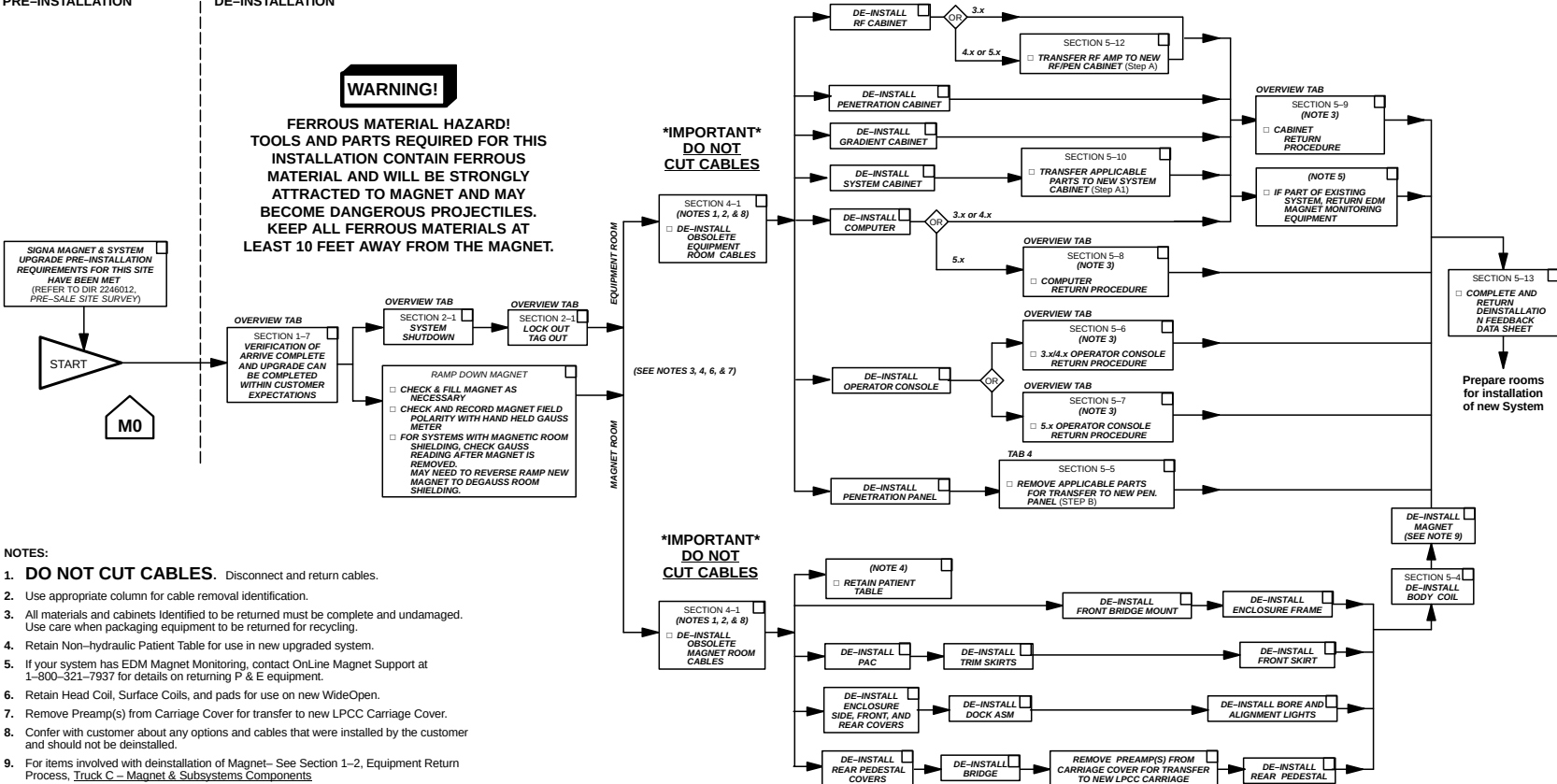
- **SIGNA 3.x**
- **SIGNA ADVANTAGE 4.x**
- **SIGNA ADVANTAGE 5.x 1.5T & 1.0T**
(Fixed-sites Only)

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

PRE-INSTALLATION

DE-INSTALLATION



- NOTES:
- DO NOT CUT CABLES.** Disconnect and return cables.
 - Use appropriate column for cable removal identification.
 - All materials and cabinets identified to be returned must be complete and undamaged. Use care when packaging equipment to be returned for recycling.
 - Retain Non-hydraulic Patient Table for use in new upgraded system.
 - If your system has EDM Magnet Monitoring, contact OnLine Magnet Support at 1-800-321-7937 for details on returning P & E equipment.
 - Retain Head Coil, Surface Coils, and pads for use on new WideOpen.
 - Remove Preamp(s) from Carriage Cover for transfer to new LPCC Carriage Cover.
 - Confer with customer about any options and cables that were installed by the customer and should not be deinstalled.
 - For items involved with deinstallation of Magnet- See Section 1-2, Equipment Return Process, Truck C – Magnet & Subsystems Components

SIGNA 3.x, 4.x, and 5.x to SIGNA WideOpen 8.x DEINSTALLATION FLOW CHART
ILLUSTRATION 3-1

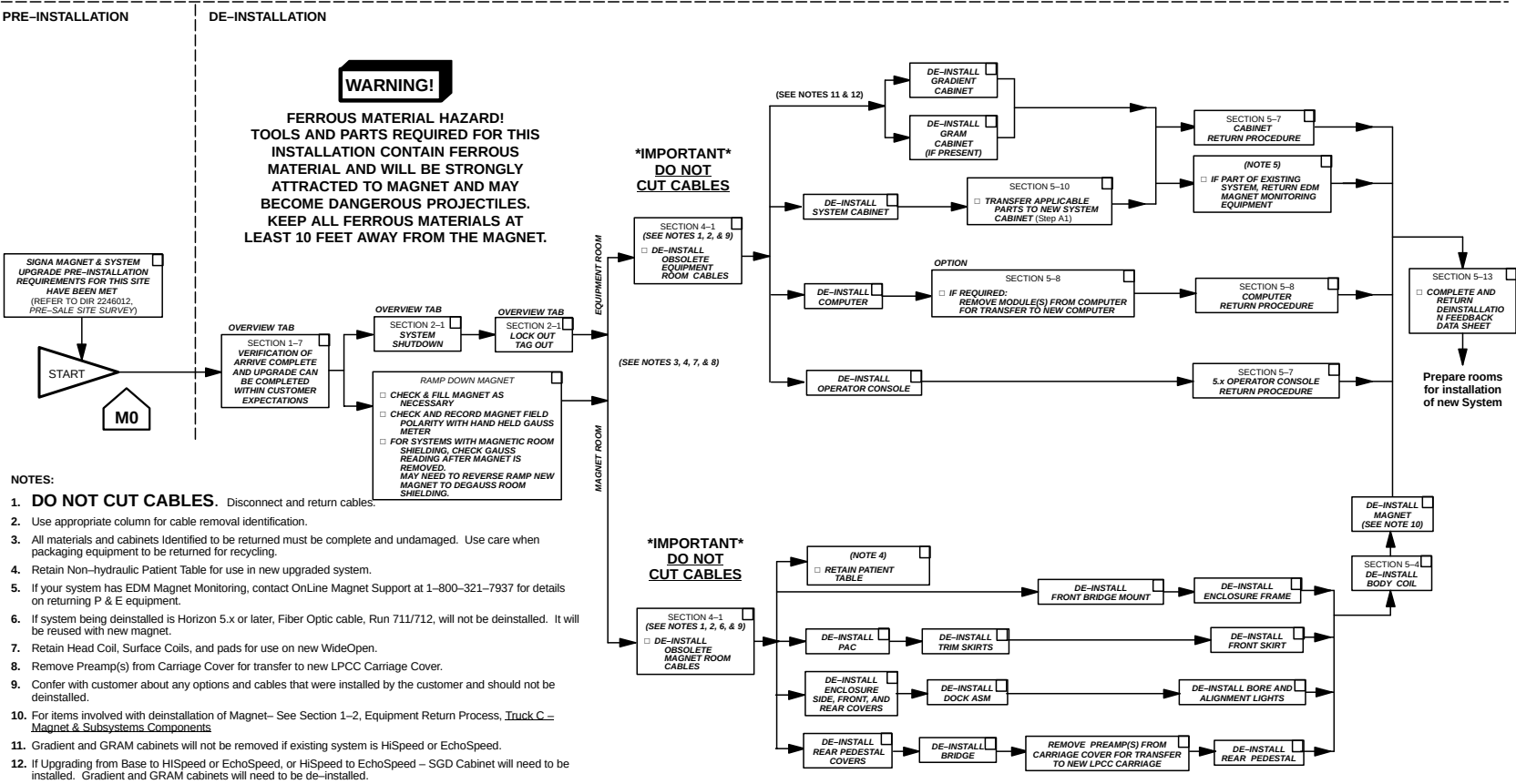
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SIGNA WideOpen 8.x UPGRADE FROM:

- **SIGNA HORIZON 5.x**
(Fixed-sites Only)

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



NOTES:

1. DO NOT CUT CABLES. Disconnect and return cables.

2. Use appropriate column for cable removal identification.

3. All materials and cabinets identified to be returned must be complete and undamaged. Use care when packaging equipment to be returned for recycling.

4. Retain Non-hydraulic Patient Table for use in new upgraded system.

5. If your system has EDM Magnet Monitoring, contact OnLine Magnet Support at 1-800-321-7937 for details on returning P & E equipment.

6. If system being deinstalled is Horizon 5.x or later, Fiber Optic cable, Run 711/712, will not be deinstalled. It will be reused with new magnet.

7. Retain Head Coil, Surface Coils, and pads for use on new WideOpen.

8. Remove Preamp(s) from Carriage Cover for transfer to new LPCC Carriage Cover.

9. Confer with customer about any options and cables that were installed by the customer and should not be deinstalled.

10. For items involved with deinstallation of Magnet- See Section 1-2, Equipment Return Process, Truck C – Magnet & Subsystems Components

11. Gradient and GRAM cabinets will not be removed if existing system is HiSpeed or EchoSpeed.

12. If Upgrading from Base to HiSpeed or EchoSpeed, or HiSpeed to EchoSpeed – SGD Cabinet will need to be installed. Gradient and GRAM cabinets will need to be de-installed.

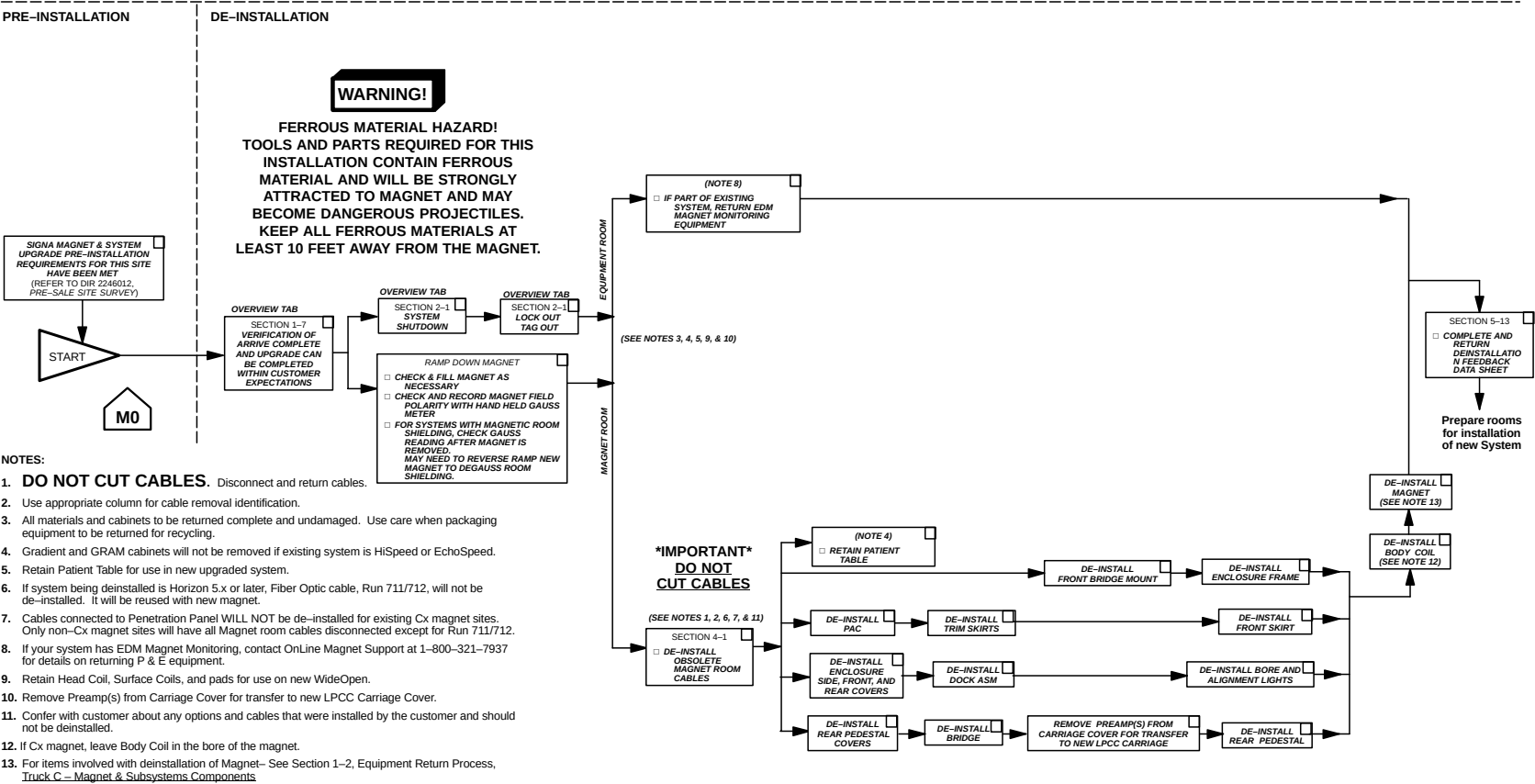
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SIGNA WideOpen 8.x UPGRADE FROM:

- **SIGNA HORIZON 8.x**
(Fixed-sites Only)

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



SECTION 4 – SYSTEM CABLES

This section provides information on removing, relabeling and installing system cables for upgrading a magnet and system to Signa MR/i or CV/i WideOpen.

4-1 CABLE REMOVAL FOR UPGRADING TO WideOpen (MR/i OR CV/i) HiSpeed OR EchoSpeed

Table 4-1 lists all cables to be removed from various early system configurations for upgrading to an 8.x WideOpen (MR/i or CV/i) HiSpeed or EchoSpeed system.

The following “Cable Removal” interconnect maps refer only to cables removed and/or relocated when upgrading:

- Signa Horizon 8.x system with Cx Magnet to Signa MR/i or CV/i Page 4-11
- Signa Horizon 8.x system to Signa MR/i or CV/i Page 4-12
- Signa Horizon 5.x system to Signa MR/i or CV/i Page 4-13
- Signa Advantage 5.4 system to Signa MR/i or CV/i Page 4-14

Signa and pre-5.4 Signa Advantage systems that are being upgraded to Signa WideOpen require many more cables to be removed or re-labeled than indicated on the cable removal maps on pages 4-11 to 4-14. Table 4-1 lists all cables to be removed for upgrading to Signa WideOpen HiSpeed or EchoSpeed from various early system configurations, including Advantage 5.4, Horizon 5.x & 8.x, and Horizon 8.x with Cx magnet.

Note

Magnet room Fiber Optic runs 711/712 will not be removed during Magnet room deinstallation for existing Horizon 5.x and 8.x systems.

The letters on the right side of table NO TAG are defined by the following cable removal explanations:

- A** – Existing system is Cx Magnet Signa Horizon 8.x with SGD Cabinet upgrading to MR/i or CV/i
- B** – Existing system is Cx Magnet Signa Horizon 8.x with Gradient/GRAM Cabinets upgrading to MR/i or CV/i
- C** – Existing system is Signa Horizon 8.x with SGD Cabinet upgrading to MR/i or CV/i
- D** – Existing system is Signa Horizon 8.x with Gradient/GRAM Cabinets upgrading to MR/i or CV/i
- E** – Existing system is Signa Horizon 5.x with SGD Cabinet upgrading to MR/i or CV/i
- F** – Existing system is Signa Horizon 5.x with Gradient/GRAM cabinets upgrading to MR/i or CV/i
- G** – Existing system is 5.4 Signa Advantage upgrading to MR/i or CV/i
- H** – Existing system is 4.5 or later but pre 5.4 Signa Advantage upgrading to MR/i or CV/i
- I** – Existing system is 4.x but pre 4.5 Signa Advantage upgrading to MR/i or CV/i
- J** – Existing system is 3.2 Performance Plus Signa upgrading to MR/i or CV/i
- K** – Existing system is Pre 3.2 Performance Plus Signa upgrading to MR/i or CV/i

4-1 CABLE REMOVAL FOR UPGRADING TO WideOpen (MR/i OR CV/i) HiSpeed OR EchoSpeed

TABLE 4-1
CABLES TO BE REMOVED AND RETURNED WHEN UPGRADING TO WideOpen

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K
001	PD1-A2-TS1-22,23,24/GND/NEUT	MR2-A12-AC IN										X	X
002	PD1-A2-TS1-32/NEUT/GND	MR2-A12-SERVICE								X	X	X	X
003	PD1-A2-TS1-1,2,3/NEUT/GND	MR3-A10-AC IN							X	X	X	X	X
005	PD1-FEEDER 16	MR3-A10-SERVICE											X
006	PD1-GND BUS	MR3-A10-GND											X
007	PD1-FEEDER 15	PP2-A6-AC IN								X	X	X	X
008	PD1-FEEDER 16	PP2-A6-SERVICE								X	X	X	X
009	PD1-A2-TS1-10,11,12/NEUT/GND	MR1-J1 (SEE NOTE 5)	X	X	X	X	X	X	X	X	X	X	X
010	PD1-A2-TS1-33/NEUT/GND	MR1-A7-J2										X	X
012	PD1-A2-TS1-21/NEUT/GND	OC1-A8-J3 (AC IN)								X	X	X	X
013	PD1-A2-TS1-33/NEUT/GND	OC1-A8-J6 (SERVICE)								X	X	X	X
014	PD1-A2-TS1-34/NEUT/GND	CC1-A20-SERVICE								X	X	X	X
015	PD1-A2-TS1-13,14,15/NEUT/GND	CC1-A20-AC IN								X	X	X	X
016	PD1-A2-TS1-16,17,18/NEUT/GND	CC2-A20-AC IN								X	X	X	X
017	PD1-A2-TS1-34/NEUT/GND	CC2-A20-SERVICE								X	X	X	X
021	PD1-A2-GND	CC1-A20-GND								X	X	X	X
022	PD1-A2-GND	CC2-A20-GND								X	X	X	X
023	PD1-GND BUS	OC1-GND					X	X	X	X	X	X	X
027	PD1-A2-TS1-32/NEUT/GND	PP2-A15-J1-SERVICE								X	X	X	X
028	PD1-A2-TS1-31/NEUT/GND	PP2-A15-J2-AC IN								X	X	X	X
029	PD1-A2-GND	MR4-GND										X	
035	PD1-A2-TS1-22,23,24/NEUT/GND	CC2-A20-AC IN								X	X	X	X
036	PD1-FEEDER 5-TS1-4,5,6 (Spect.)	MR6 (Spectroscopy Only)							X	X	X		
038	PD1-GND BUS	MR1-A7 GND							X				
043	MR7-A4-J7	PD1-A2											
045	PP1-GND (See Note 2)	FAN MODULE GND (Note 2)			X	X	X	X					
050	OC1-AC-IN	PD1-TS1-13,14,15,NUE,GND						X	X				
064	MR2-A12-J1-AC OUT	CC1						X	X				
200	MR2-A11-J19	MR3-A9-J1								X	X	X	X
201	MR2-A11-J20	MR3-A9-J2								X	X	X	X
202	MR2-A11-J21	MR3-A9-J3								X	X	X	X
203	MR2-A11-J22	MR3-A9-J4								X	X	X	X
204	MG3-A2	MR3-TS1-1/2											X
205	MG3-A3	MR3-TS1-3/4											X
206	MG3-A4	MR3-TS1-5/6											X
210	PP1-J16	MR2-A11-J9											X
211	MR2-A11-J23	PP2-A5-J1											X
212	PP1-J19	MR2-A11-J25										X	X
213	PP1-J17	MR2-A11-J10										X	X
214	PP1-J21	MR2-A11-J27										X	X

4-1 CABLE REMOVAL FOR UPGRADING TO WideOpen (MR/i OR CV/i) HiSpeed OR EchoSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K
215	PP1-J22	MR2-A11-J28										X	X
216	PP1-J23	MR2-A11-J29										X	X
217	PP1-J26	MR2-A11-J32										X	X
218	PP1-J25	MR2-A11-J31										X	X
219	PP1-J24	MR2-A11-J30										X	X
220	PP1-J27	MR2-A11-J33										X	X
221	PP1-J28	MR2-A11-J34										X	X
223	OC1-A8-J13	MR2-A1-J36								X	X	X	X
224	CC1-A21-J19	MR1-A7-J8								X	X	X	X
225	MR1-A7-J10	MR2-A11-J38										X	X
226	MR1-A7-J5	MR2-A11-J14								X	X	X	X
228	CC1-A21-J10	MR2-A11-J17								X	X	X	X
230	CC1-A21-J30	MR2-A11-J39										X	X
232	PP1-J6	MR2-A11-J4										X	X
234	MR2-A11-J12	PP2-A5-J2											X
237	PP1-J15	MR1-A11-J7											X
238	CC1-A21-J14	OC1-A8-J8								X	X	X	X
239	CC1-A21-J15	OC1-A8-J10								X	X	X	X
244	CC1-A21-J13	BTI-HOST								X		X	X
248	CC2-A21-J1	OC1-A9-J1								X	X	X	X
249	CC2-A21-J22	OC1-A8-J12								X	X	X	X
256	PP1-J12	OC1-A8-J11								X	X	X	X
257	PP1-J44 (See Note 1)	MG3-A3-A1-J1 (See Note 1)			X	X	X	X	X	X	X	X	X
258	PP1-J4 (See Note 1)	MG3-A17-J1 (See Note 1)			X	X	X	X	X	X	X	X	X
259	PP1-J5	SURFACE COIL											X
261	PP1-J6	MG2-A11-A1-J1							X	X	X	X	X
262	PP1-J8	MG3-A3-A5-OUT	X	X	X	X	X	X	X	X	X	X	X
263	PP1-J19	MG2-A4-J1										X	X
364	PP1-J24	MG2-A1-A5-J5										X	X
365	PP1-J25	MG2-A1-A5-J4										X	X
266	PP1-J26	MG2-A1-A5-J6										X	X
267	PP1-J23	MG2-A1-A5-J3										X	X
268	PP1-J22	MG2-A1-A5-J1										X	X
269	PP1-J21	MG2-A1-A5-J2										X	X
270	PP1-J16	EQ1											X
273	PP1-J17	O2 MONITOR (OM1)										X	X
274	CC1-A21-J20	MR1-A7-J7								X	X	X	X
275	PP1-J27	MG3-A1-J4										X	X
280	PP1-J15	PP2-A1-J2								X	X	X	X
281	PP1-J11	PP2-A1-J3								X	X	X	X
282	PP1-J12	MG3-A2-J6			X	X	X	X	X	X	X	X	X
286	CC1-A21-J1	PD1-A1-A3-J2										X	X
287	CC2-A21-J15	CC1-A21-J2								X	X	X	X
297	PP1-J18	EO1											

4-1 CABLE REMOVAL FOR UPGRADING TO WideOpen (MR/i OR CV/i) HiSpeed OR EchoSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K
300	PP1-J15	MG2-A15	X	X	X	X	X	X	X	X	X	X	X
301	CC2-A21-J20	CC1-A21-J28								X	X	X	X
302	CC2-A21-J21	CC1-A21-J29								X	X	X	X
318	MG3-A7	PP1-J11			X	X	X	X	X	X	X	X	X
319	MR1-A7-J4	MR2-A11-J15											X
320	MR1-A7-J6	MR2-A11-J8											X
321	MR2-A11-J37	MR1-A7-JP										X	X
322	MR2-A11-J11	PP2-A6-J18								X	X	X	X
324	MR1-A7-J17	MR2-A11-J49											X
325	CC1-A21-J27	CC2-A21-J19								X	X	X	X
327	CC1-A21-J25	CC2-A21-J17								X	X	X	X
328	PP1-J3	MR1-A7-J18											X
329	PP1-J3	MG2-A12-A2-J7											X
340	MG3-A3-A1-J4	MG2-A12-A2-J1										X	X
341	MG3-A3-A1-J3	MG2-A12-A2-J2										X	X
344	MG2-A12-A2-J7	MG2-A12-A2-J6										X	
346	MR2-A11-J41	CM1-FRONT								X	X	X	X
347	MR2-A11-J42	CM1-BACK								X	X	X	X
354	CM1	ST1								X	X	X	X
370	PP1-A9-J46	MR1-A7-J19									X	X	
384	PP1-J20	PP2-A15-J17							X	X	X	X	X
385	PP1-J20	MG2-A30	X	X	X	X	X	X	X	X	X	X	X
386	PP1-J47	PP2-A15-J21							X	X	X	X	X
387	PP1-J47	MG2-A29			X	X	X	X	X	X	X	X	X
388	PP1-J11	PP2-A15-J22							X	X	X	X	X
389	PP1-J15	PP2-A15-J20							X	X	X	X	X
390	MR2-A11-J11	PP2-A15-J3										X	X
391	MR2-A11-J23	PP2-A15-J18										X	X
392	MR2-A11-J12	PP2-A15-J19										X	X
402	PP1-J28	MG2-A31-J1										X	
404	MG3-A17-J1	MG2-A16-A5-J6	X	X	X	X	X	X	X	X	X		
405	MR3-A9-TS1	MG3-TS1-1,2							X	X	X	X	
406	MR3-A9-TS1	MG3-TS1-3,4							X	X	X	X	
407	MR3-A9-TS1	MG3-TS1-5,6							X	X	X	X	
408	PD1-A1-A3-J3	PP1-J16							X	X	X	X	
409	PP1-J16	MG2-A12-A1-J7							X	X	X	X	
410	PP1-J14	PP2-A15-J23									X	X	
411	PP1-J14	MG3-A13-J1									X	X	
412	PP1-J51	MG3-A32-J1							X	X	X	X	
413	PP1-J51	PP2-A15-J28(A16-J9)							X	X	X	X	X
414	MG3-A35	MG3-A2-J9	X	X	X	X	X	X	X	X	X		
415	PP1-A8-J45	MG3-A33-J7									X	X	
416	MR1-A7-J21	PP1-A8-J45									X	X	
417	PP2-A15-J19	MR2-A11-J32								X	X		
418	MR2-A11-J39	PP2-A15-J3									X		

4-1 CABLE REMOVAL FOR UPGRADING TO WideOpen (MR/i OR CV/i) HiSpeed OR EchoSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K
419	MR2-A11-J26	PP1-A10-J1								X	X		
420	MG3-A33-J1	PP1-A10-J2								X	X		
421	MG2-A33-J3	MG3-A31-J1	X	X	X	X	X	X	X	X	X		
422	MG2-A33-J2	MG3-A1-J4	X	X	X	X	X	X	X	X	X		
424	PP1-A10-J5/J6	MG2-A33-J11/J12								X	X		
425	MR2-A11-J12	PP1-J6									X		
426	MR1-A7-J9	MR3-A9-J5								X	X		
427	MR2-A11-J17	CC1-A21-J13								X	X		
428	MR2-A11-J44	CC1-A21-J35								X	X		
429	BT1	CC1-A21-J25								X	X		
430	MR2-A11-J24	CC1-A21-J26								X	X		
431	MR2-A11-J25	CC1-A21-J27								X	X		
433	OC1-A9-J1	CC1-A21-J10								X	X		
434	CC1-A8-J12	CC1-A21-J29								X	X		
435	MR2-A11-J13	PP1-A10-J3								X	X		
436	MG2-A12-A1-J1	MG3-A33-J3							X	X	X		
437	MG2-A12-A1-J2	MG3-A33-J4							X	X	X		
439	MG3-A3-J4	MG3-A33-J1							X	X	X		
440	MG3-A3-J3	MG3-A34-J2							X	X	X		
441	MG3-A33-J7	MG3-A34-J8							X	X	X		
446	PP1-A9-J46	MG2-A12-A13-J5									X	X	
447	PP1-A9-J46	MG2-A12-A13-J3									X	X	
448	MR2-A11-J44	CC1-A21-J1								X			
449	MR2-A11-J24	CC1-A21-J34								X			
450	MR2-A11-J25	CC1-A21-J35								X			
453	PP1-J50	MG2-A11-A1-J4								X	X		
454	MR2-A11-J34	PP1-J50								X	X		
470	MR1-A7-J17	PP1-A11-J71							X	X			
474	PP1-A11-J72	MG3-A12-A1-J3							X	X			
475	PP1-A11-J73	MG3-A12-A1-J4							X	X			
476	PP1-A11-J74	MG3-A33-J7							X	X			
477	PP1-A11-J75	MG3-A12-A1-J5							X	X			
478	PP1-A11-J76	MG3-A12-A1-J6							X	X			
485	MG3-A17-J2	MG2-A16-A3-OUT			X	X	X	X	X	X	X		
491	PP1-J80	MG3-A36-J7			X	X	X	X	X	X	X		
492	PP1-J81	MG3-A36-J8			X	X	X	X	X	X	X		
493	PP1-J82	MG3-A36-J9			X	X	X	X	X	X	X		
494	PP1-A14-J87	MG3-A36-J6			X	X	X	X	X	X	X		
495	PP1-A15-J78	MG3-A36-J10			X	X	X	X	X	X	X		
500	PP1-J39	MG2-A14											X
519	PP1-J39	PP2-A1-J27											X
520	PP1-J40	PP2-A1-A2-1											X
521	PP1-J41	PP2-A1-A2-2											X
522	PP1-J40	MG2-A12-J3											X
523	PP1-J41	MG2-A12-J4											X

4-1 CABLE REMOVAL FOR UPGRADING TO WideOpen (MR/i OR CV/i) HiSpeed OR EchoSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K
601	MS1 (RED)	MS6 (RED)	X	X	X	X	X	X	X	X	X	X	X
602	MS1 (BLK)	MS6 (BLK)	X	X	X	X	X	X	X	X	X	X	X
603	MS1-A13-A1	MS7	X	X	X	X	X	X	X	X	X	X	X
604	MS1-A13-A1	MS6/MS7	X	X	X	X	X	X	X	X	X	X	X
605	PP1-J10 (Magnet Room)	MS1-A3-A1 (Magnet Room)	X	X	X	X	X	X	X	X	X	X	X
605	PP1-J10 (Equipment Room)	MR1-J11 (Equipment Room)	X	X	X	X	X	X	X	X	X	X	X
606	MS4	MS1-A3-A1	X	X	X	X	X	X	X	X	X	X	X
621	MS5-A1 (Gas Line)	MS1-A2 (Gas Line)	X	X	X	X	X	X	X	X	X	X	X
622	MS5-A1 (Gas Line)	MS1-A2 (Gas Line)	X	X	X	X	X	X	X	X	X	X	X
623	MS5-A1	PP1-F1,F2,F3,F4	X	X	X	X	X	X	X	X	X	X	X
624	MS1-A2	PP1-F1,F2,F3,F4	X	X	X	X	X	X	X	X	X	X	X
626	MS5-A1	BALZER REMOTE			X	X	X	X	X	X	X		
700	PP1-A14-J86	MG3-A17-J2			X	X	X	X	X				
704	PP1-J12	OC1-A10-J9					X	X	X				
707	MR2-A11-J26-ESTOP	OC1-J8					X	X	X				
708	MR2-A11-J16	MR1-A7-J27							X				
709	MR1-A7-J26	MR3-A9-J1							X				
710	MR2-A11-J17	MR3-A9-J2							X				
711	PP1-A17-A1-J91	MG2-A33							X				
712	PP1-A17-A1-J91	MG2-A11-A1							X				
715	PP1-J14	MG2-A33-J1			X	X	X	X	X	X	X		
716	PP1-J89	MG2-A11-A1-J2			X	X	X	X	X	X	X		
718	MR2-A11-J29	OC1-A10-J4					X	X	X				
719	CC1-A1-A6-J1	MR2-A15-A3-J1					X	X	X				
727	PP1-A16-J77	MG3-A36-J5			X	X	X	X					
728	MG3-A36-J11	MG2-A16-OUT-J1	X	X	X	X	X	X	X	X	X		
729	MG3-A36-J12	MG2-A16-OUT-J2	X	X	X	X	X	X	X	X	X		
730	MG3-A36-J13	MG2-A16-OUT-J3	X	X	X	X	X	X	X	X	X		
731	MG3-A36-J14	MG2-A16-OUT-J4	X	X	X	X	X	X	X	X	X		
732	MG3-A36-J15	MG2-A16-OUT-J5	X	X	X	X	X	X	X	X	X		
733	MG3-A36-J16	MG2-A16-OUT-J6	X	X	X	X	X	X	X	X	X		
734	MG3-A36-J17	MG2-A16-OUT-J7	X	X	X	X	X	X	X	X	X		
735	MG3-A36-J18	MG2-A16-OUT-J8	X	X	X	X	X	X	X	X	X		
736	MG3-A36-J19	MG2-A16-OUT-J9	X	X	X	X	X	X	X	X	X		
743	MG3-A3-J3	MG2-A12-J2	X	X	X	X	X	X					
744	MG3-A3-J4	MG2-A12-J1	X	X	X	X	X	X					
746	PP1-J44 (See Note 1)	MG3-A3-A1-J1 (See Note 1)			X	X	X	X	X	X	X		
747	PP1-J4 (See Note 1)	MG3-A17-J1 (See Note 1)			X	X	X	X	X	X	X		
749	MG2-A36	MG2-A33-J10	X	X	X	X	X	X	X				
753	MR7-A4-J1	MG3-A11-J5											
754	MR7-A4-J2	MG3-A11-J4											
755	MR7-A4-J3	MG3-A11-J3											
756	MR7-A4-J4	MG3-A11-J8											

4-1 CABLE REMOVAL FOR UPGRADING TO WideOpen (MR/i OR CV/i) HiSpeed OR EchoSpeed

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K
757	MR7-A4-J5	MG3-A11-J7											
758	MR7-A4-J6	MG3-A11-J6											
759	MR3-A11-J12	MR7-A4-J8											
760	MR3-A11-J11	MR7-A4-J9											
761	MR3-A11-J10	MR7-A4-J10											
762	MG3-A32-TS1/2 (See Note 3)	PP1-A7-1/2 (See Note 3)			X	X	X	X					
763	MG3-A32-TS3/4 (See Note 3)	PP1-A7-3/4 (See Note 3)			X	X	X	X					
764	MG3-A32-TS5/6 (See Note 3)	PP1-A7-5/6 (See Note 3)			X	X	X	X					
767	MR1-A7-J37 (See Note 4)	PP1-J51 (See Note 4)						X					
778	MG2-A12-A13-J3	PP1-A11-J72			X	X	X	X					
779	MG2-A12-A13-J4	PP1-A11-J73			X	X	X	X					
780	MG3-A3-J6	PP1-A11-J74			X	X	X	X					
781	MG2-A12-A12-J5	PP1-A11-J75			X	X	X	X					
782	MG2-A12-A12-J6	PP1-A11-J76			X	X	X	X					
784	PP1-F5/F6	MG6-J1			X	X	X	X					
786	PP1-F7/F8	MG6-J2			X	X	X	X					
800	OC1-J1	CC1-A2-J1 (MUX)					X	X	X				
801	OC1-J10	CC1-A2-J2 (SECUR. KEY)					X	X	X				
802	OC1-J12	CC1-A1-J9 (PLASMA)					X	X	X				
803	OC1-J13	CC1-A1-A10 (VIDEO)					X	X	X				
804	OC1-J3	CC1-A1-A1 (BOOT)					X	X	X				

Notes:

- 200 series cables are 1.5T specific. 700 series cables are 1.0T specific. These cable will have labels added after they are routed and terminated.
- Only present if system is at Horizon
- Only the magnet room half is removed and replaced. Only present if system is at Horizon.
- Run 767 is only found on a few early Horizon 5.x (5.5) systems.
- For new RF/PEN cabinet, Run 009 requires 3 feet of slack for service loop. The module that the cable connects to moves forward for servicing. If cable is not long enough for service loop, it should be replaced by a new longer cable.

4-2 RE-LABELING AND RE-ROUTING OF EXISTING PRE-HORIZON CABLES TO 8.x WideOpen

Several existing cables need to be re-labeled and then re-routed per the new labels. Refer to Table 4-2.

Runs 706, 772, 773, and 774 are only re-labeled for Advantage 5.4 systems upgraded to 8.x WideOpen. All pre-5.4 system upgrades are provided new cables for Runs 706, 772, 773, and 774.

TABLE 4-2
PRE-HORIZON CABLES RE-LABELED AND RE-LOCATED FOR UPGRADE TO WideOpen

OLD RUN	SYSTEM TYPE	ONE END		OTHER END	
		OLD LABEL	NEW	OLD LABEL	NEW LABEL
207	3.x 4.x	RUN #207 MR2-A11-J6 46-243775G411	RUN #701 MR2-A11-J15 46-243775G807	RUN #207 DOOR SWITCH 46-243775G411	RUN #701 RF DOOR SWITCH 46-243775G807
227	3.x 4.x	RUN #227 MR2-A11-J16 46-243775G464	RUN #703 MR2-A11-J23 46-243775G809	RUN #227 PD1-A1-A2-J2 46-243775G464	RUN #703 PD1-A1-A2-J2 46-243775G809
387	3.x	RUN #387 IS NOT LABELED USE LABELS PROVIDED	RUN #387 PP1-J47 46-243775G531	RUN #387 IS NOT LABELED USE LABELS PROVIDED	RUN #387 MG2-A29-P1 46-243775G531
432	4.x 5.x	RUN #432 PD1-A1-A3-J2 46-243775G676	RUN #706 PD1-A1-A3-J2 46-243775G808	RUN #432 MR2-A11-J45 46-243775G676	RUN #706 MR2-A11-J14 46-243775G808
705	5.x	RUN #705 MR2-A11-J25 46-243775G663	RUN #774 MR2-A11-J25 46-243775G811	RUN #705 PP2-A15-J19 46-243775G663	RUN #774 MR1-A7-A32 46-243775G811
713	5.x	RUN #713 PP2-A15-J23 46-243775G676	RUN #772 MR1-A7-J40 46-243775G808	RUN #713 PP1-J14 46-243775G676	RUN #772 PP1-J14 46-243775G808
714	5.x	RUN #714 PP2-A5-J27 46-243775G676	RUN #773 MR1-A7-J35 46-243775G808	RUN #714 PP1-J89 46-243775G676	RUN #773 PP1-J89 46-243775G808

4-3 RE-LABELING & RE-ROUTING HORIZON 5.x CABLES TO 8.x WideOpen WITH NEW SGD

Several existing Horizon 5.x cables need to be re-labeled and then re-routed per the new labels. Refer to Table 4-3.

Runs 041, 042, 709, and 710 are re-labeled only for Horizon 5.x systems upgraded to 8.x with SGD. All pre-5.x system upgrades are provided new cables for Runs 041, 042, 709, and 710

TABLE 4-3
HORIZON 5.x CABLES RE-LABELED AND RE-LOCATED FOR UPGRADE 8.x WideOpen

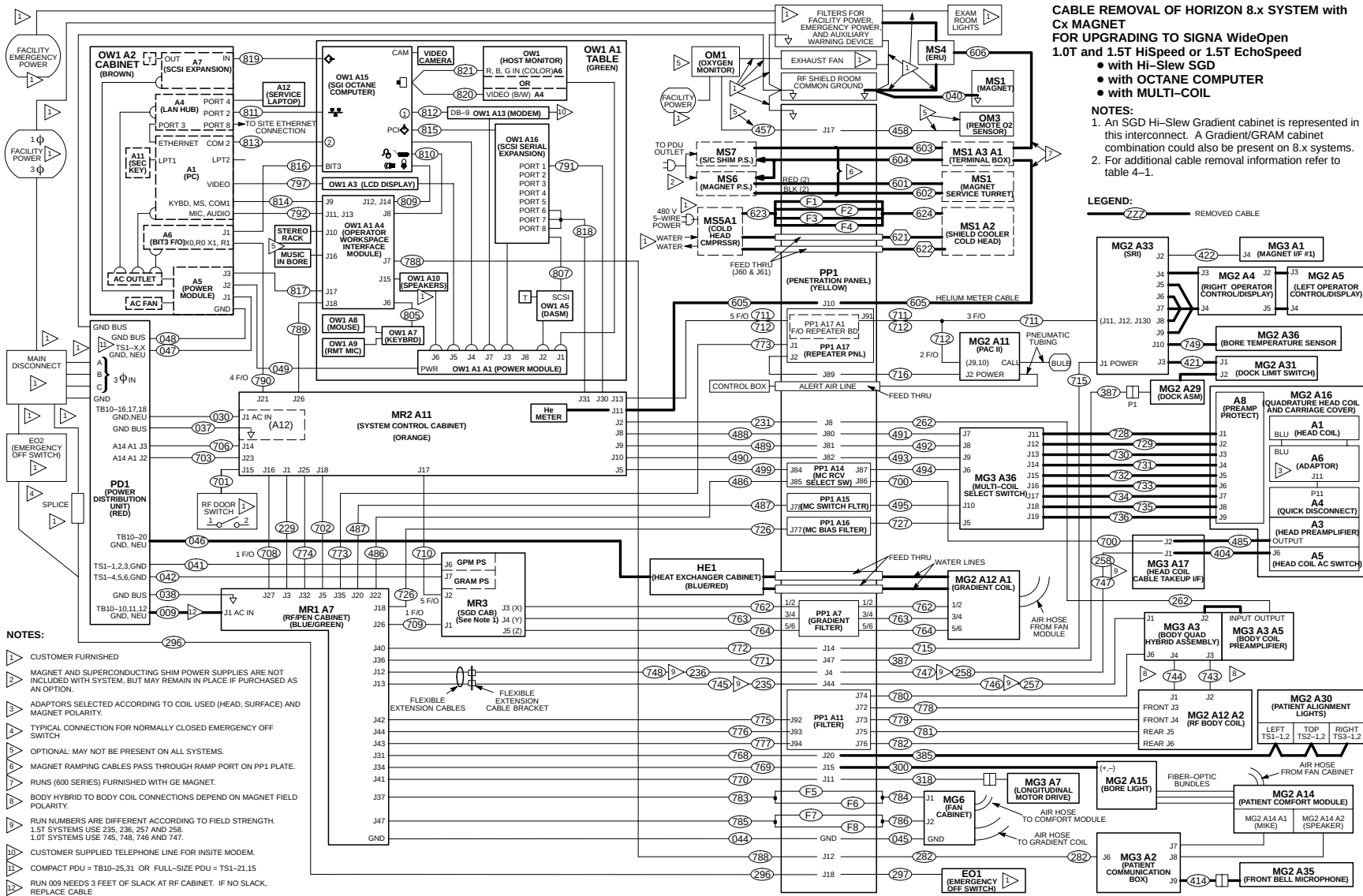
OLD RUN	ONE END		OTHER END	
	OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
041	RUN #041 MR3-A11-J9 46-271363G3	RUN #041 MR3CABI/FJ6 46-271363G5	RUN #041 PD1-A2 46-271363G3	RUN #041 PD1-A2 46-271363G5
042	RUN #042T MR3-A12 46-271363G604	RUN #042 MR3CABI/FJ7 46-271363G6	RUN #042T PD1-A2 46-272363G604	RUN #042 PD1-A2 46-271363G6
709	RUN #709TR-1 MR3-A11-J1 46-317077G802	RUN #709TR-1 MR3CABI/FJ2 46-317077G803	RUN #709TR-1 MR1-A7-RX 46-317077G802	RUN #709TR-1 MR1-A7-RX 46-317077G803
710	RUN #710TR-1 MR3-A11-J2 46-317085G602	RUN #710TR-1 MR3CABI/FJ1 46-317085G603	RUN #710TR-1 MR2-A11-J17 46-317085G602	RUN #710TR-1 MR2-A11-J17 46-317085G603
710-1	RUN #710-1	RUN #710-1 MR3GIPJ2	NO CHANGE	NO CHANGE
710-2	RUN #710-2	RUN #710-2 MR3GIPJ3	NO CHANGE	NO CHANGE
710-3	RUN #710-3	RUN #710-3 MR3GIPJ4	NO CHANGE	NO CHANGE
710-4	RUN #710-4	RUN #710-4 MR3GIPJ1	NO CHANGE	NO CHANGE
710-5	RUN #710-5	RUN #710-5 MR3GIPJ5	NO CHANGE	NO CHANGE

4-4 LABELS ADDED TO NEW MAGNET ROOM CABLES (Non-Cx Magnet Only)

The following table defines the additional cables which need to be relabeled when upgrading to a LCC WideOpen magnet, unless the magnet being replaced is a Cx magnet. **In the case of a Cx magnet being replaced by a LCC magnet, the cables in the following table do not need to be relabeled.**

TABLE 4-4
MAGNET ROOM LABELS TO BE RE-LABELED

OLD RUN	ONE END		OTHER END	
	OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
257 (1.5T)	NONE	RUN #257 MG3 A3 J1 46-	NONE	RUN #257 PP1-J44 46-
258 (1.5T)	NONE	RUN #258 MG3 A17 J1 46-	NONE	RUN #257 PP1-J4 46-
746 (1.0T)	NONE	RUN #746 MG3 A3 J1 46-	NONE	RUN #746 PP1-J44 46-
747 (1.0T)	NONE	RUN #747 MG3 A17 J1 46-	NONE	RUN #747 PP1-J4 46-
711/712	NONE	RUN #711/712 MG2 A33 46-328079G6	NONE	RUN #711/712 PP1-J91 46-328079G6
045	NONE	RUN #045 PP1 GND 46-	NONE	RUN #045 MG6 GND 46-



CABLE REMOVAL OF HORIZON 8.x SYSTEM with Cx MAGNET
FOR UPGRADING TO SIGNA WideOpen
1.0T and 1.5T HiSpeed or 1.5T EchoSpeed

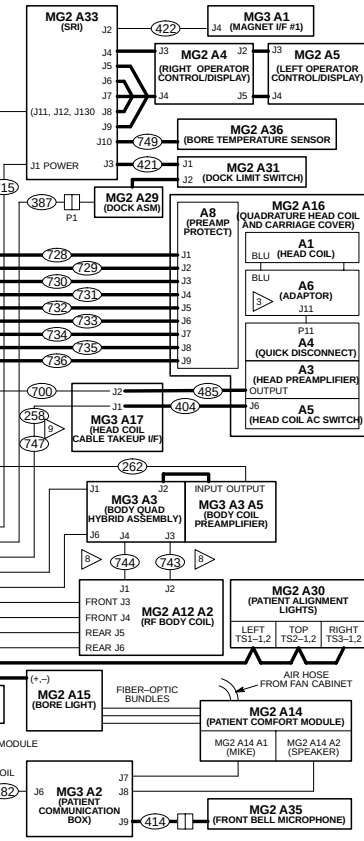
- with Hi-Slew SGD
- with OCTANE COMPUTER
- with MULTI-COIL

NOTES:

- An SGD Hi-Slew Gradient cabinet is represented in this interconnect. A Gradient/GRAM cabinet combination could also be present on 8.x systems.
- For additional cable removal information refer to table 4-1.

LEGEND:

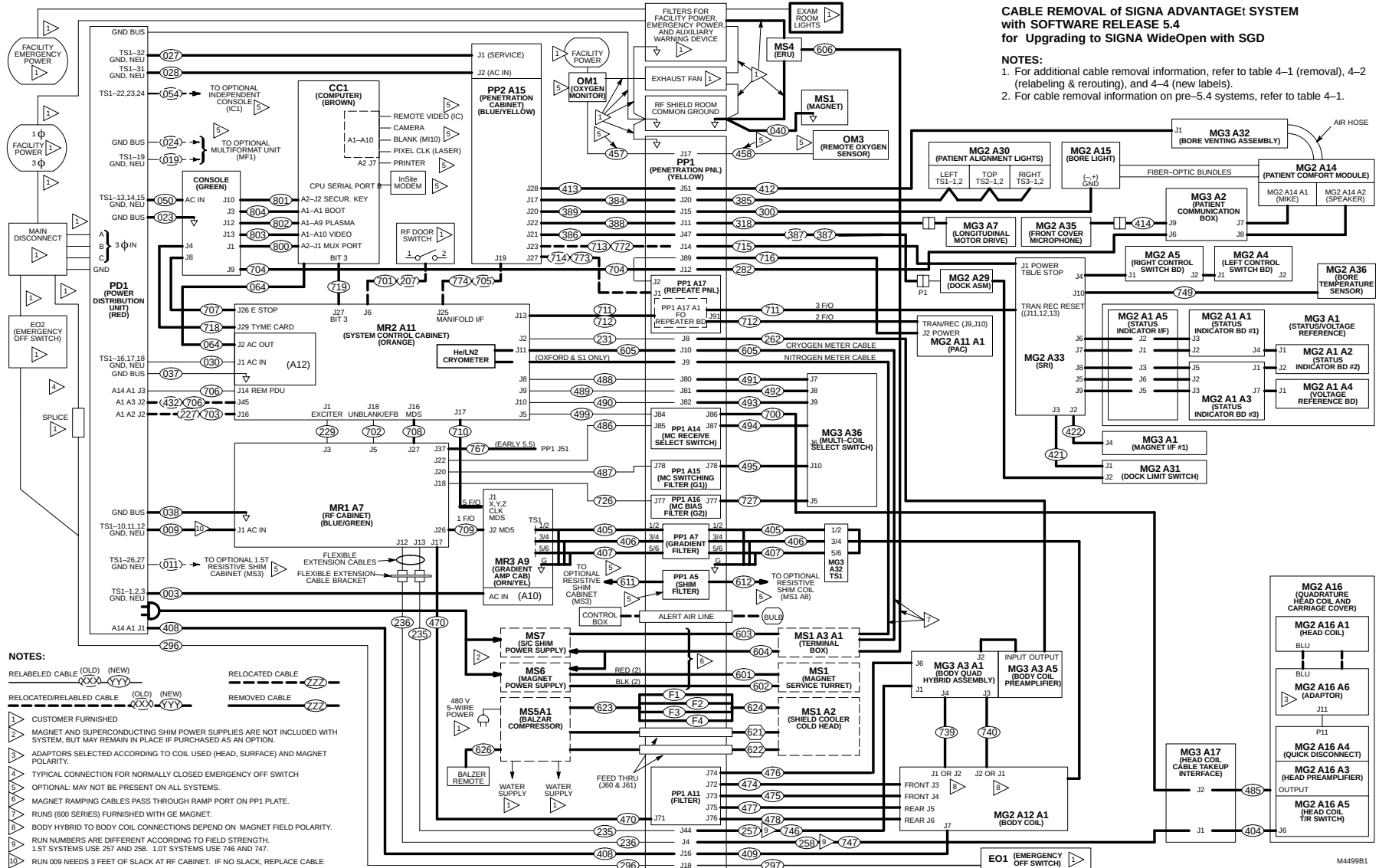
777 — REMOVED CABLE



CABLE REMOVAL of SIGNA ADVANTAGE! SYSTEM **with SOFTWARE RELEASE 5.4** **for Upgrading to SIGNA WideOpen with SGD**

NOTES:

1. For additional cable removal information, refer to table 4-1 (removal), 4-2 (relabeling & rerouting), and 4-4 (new labels).
2. For cable removal information on pre-5.4 systems, refer to table 4-1.



SECTION 5 – DEINSTALLATION

5-1 DEINSTALL MAGNET ENCLOSURE

The following are basic guidelines for all deinstallations:

- A. **DO NOT CUT CABLES.** Cables are to be disconnected and returned intact.
- B. All fastening hardware needs to be returned for reuse. In all cases, after disassembly, loosely re-install all fastening hardware in mounting holes or threaded holes where they were originally located.
- C. Retain on site the Head Coil, all Surface Coils, and pads for reuse on new WideOpen system.

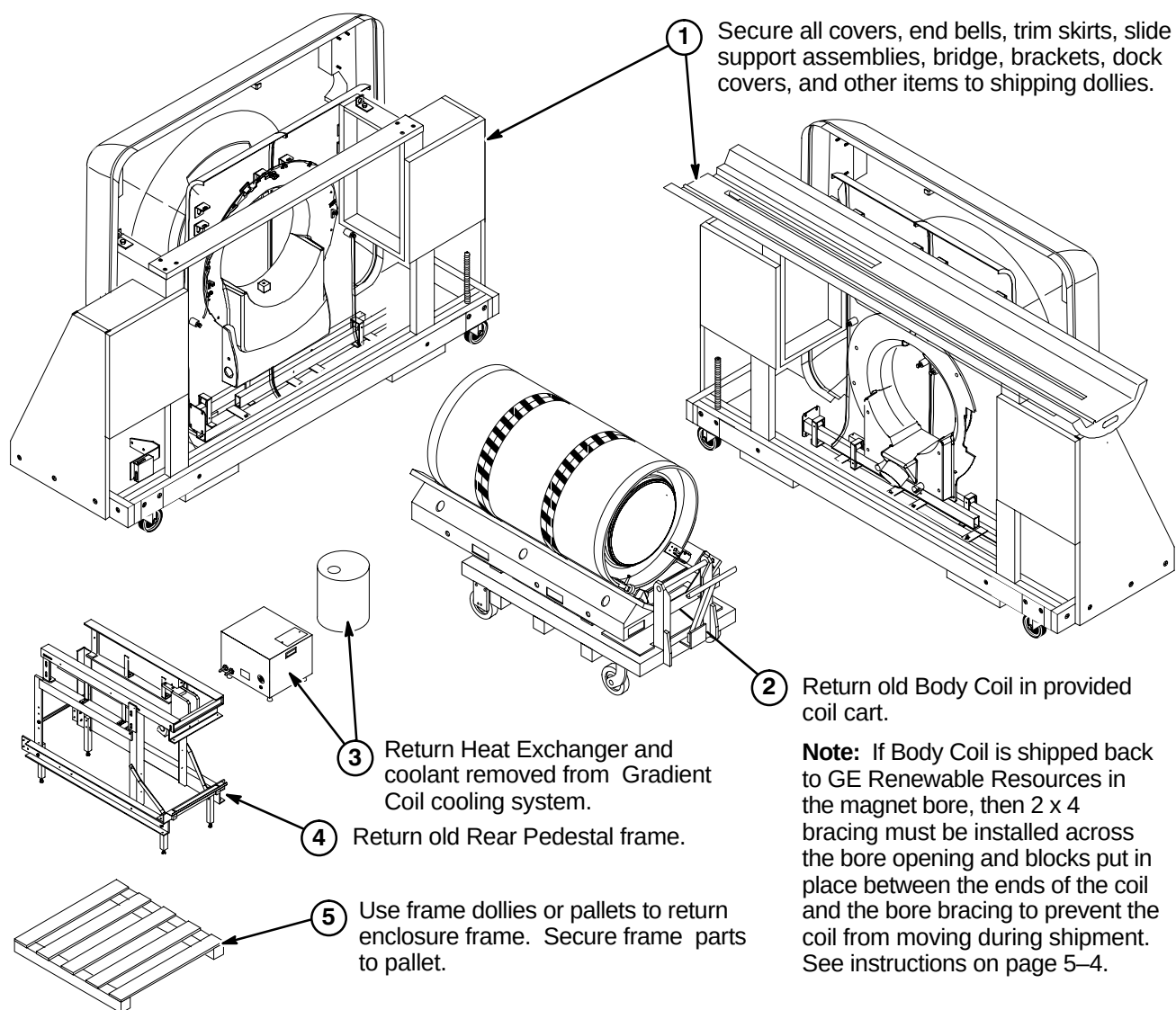
5-2 DEINSTALLATION FEEDBACK FORM

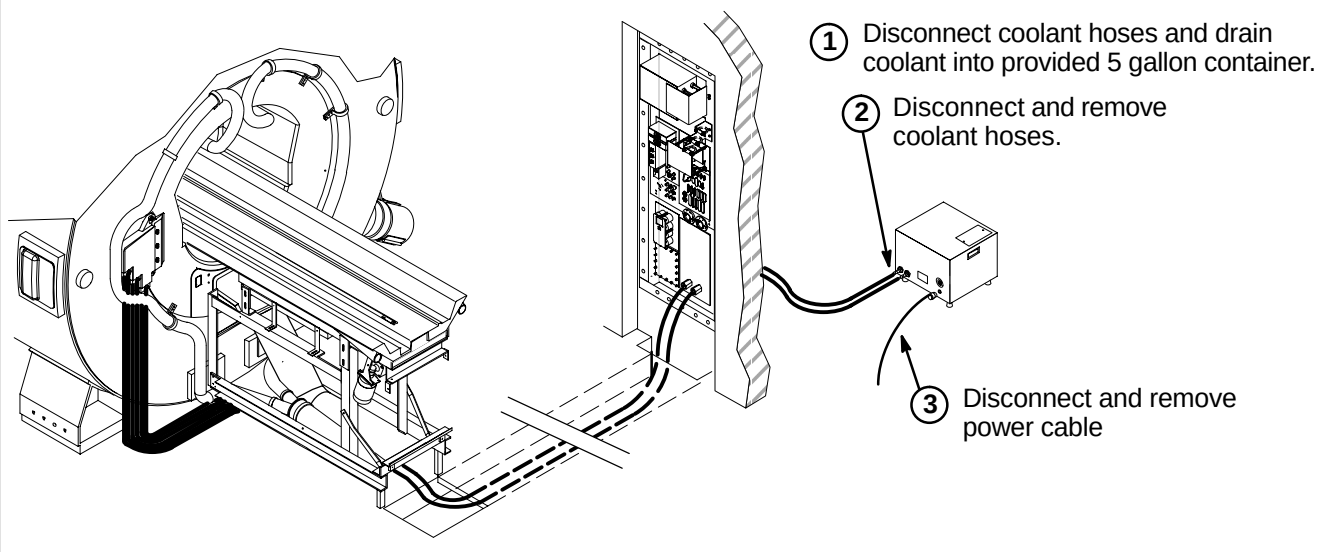
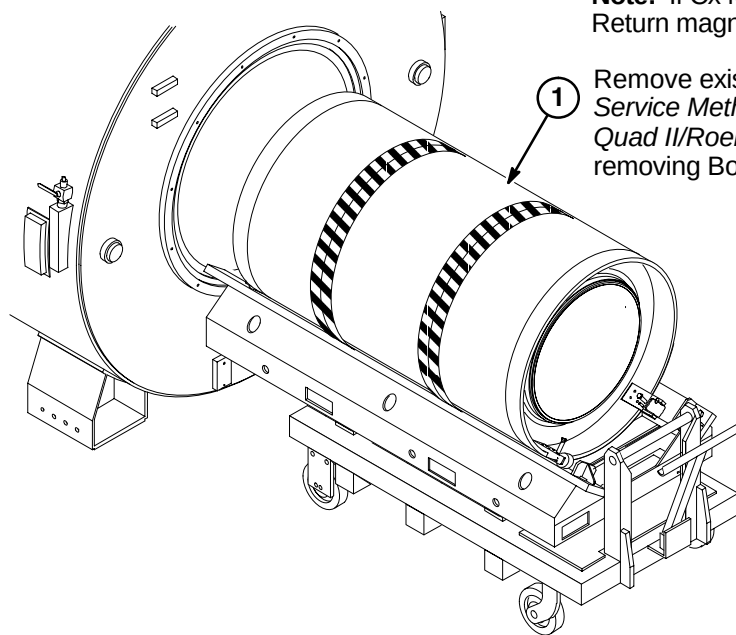
Located on the last page of this section is the Deinstallation Feedback Data Sheet. Fill out this form and send it to the address located on the bottom of the sheet. Any and all information provided will help improve the deinstallation process.

5-3 RETURN OF DEINSTALLED MATERIAL

□ ITEMS TO BE RETURNED FOR RECYCLING

Note: All items de-installed during this upgrade are to be returned for recycling. Secure as many of the items as possible to the two shipping dollies provided for this purpose. Shown below is an example of how the dollies might look when loaded. (not all items are shown)



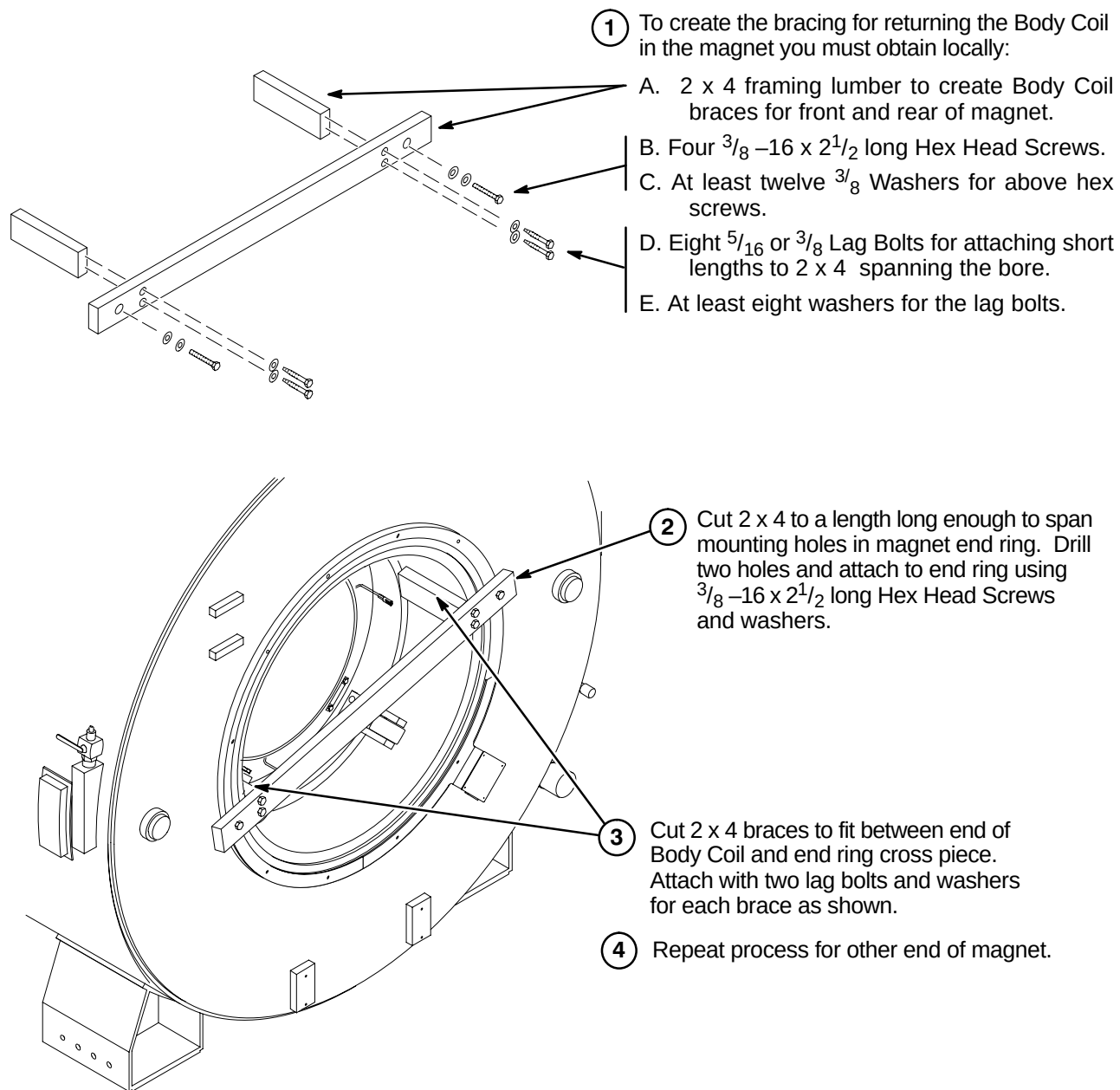
5-4 DEINSTALL BODY COIL (EXCEPT FOR Cx MAGNETS)**□ DRAIN AND REMOVE HEAT EXCHANGER (Horizon Systems)****□ REMOVE BODY COIL**

Note: If Cx Magnet, do not remove Body Coil. Return magnet with Body Coil intact.

Note: If Body Coil is shipped back to GE Renewable Resources in the magnet bore, then 2 x 4 bracing must be installed across the bore opening and blocks put in place between the ends of the coil and the bore bracing to prevent the coil from moving during shipment. See instructions on page 5-4.

5-4 DEINSTALL BODY COIL (continued)

■ BRACING BODY COIL WITHIN MAGNET BORE (Alternative)



5-5 DEINSTALL PENETRATION PANEL**■ TRANSFER OF PENETRATION PANEL COMPONENTS**

CAUTION: All cable connections MUST BE LABELLED before disconnect.

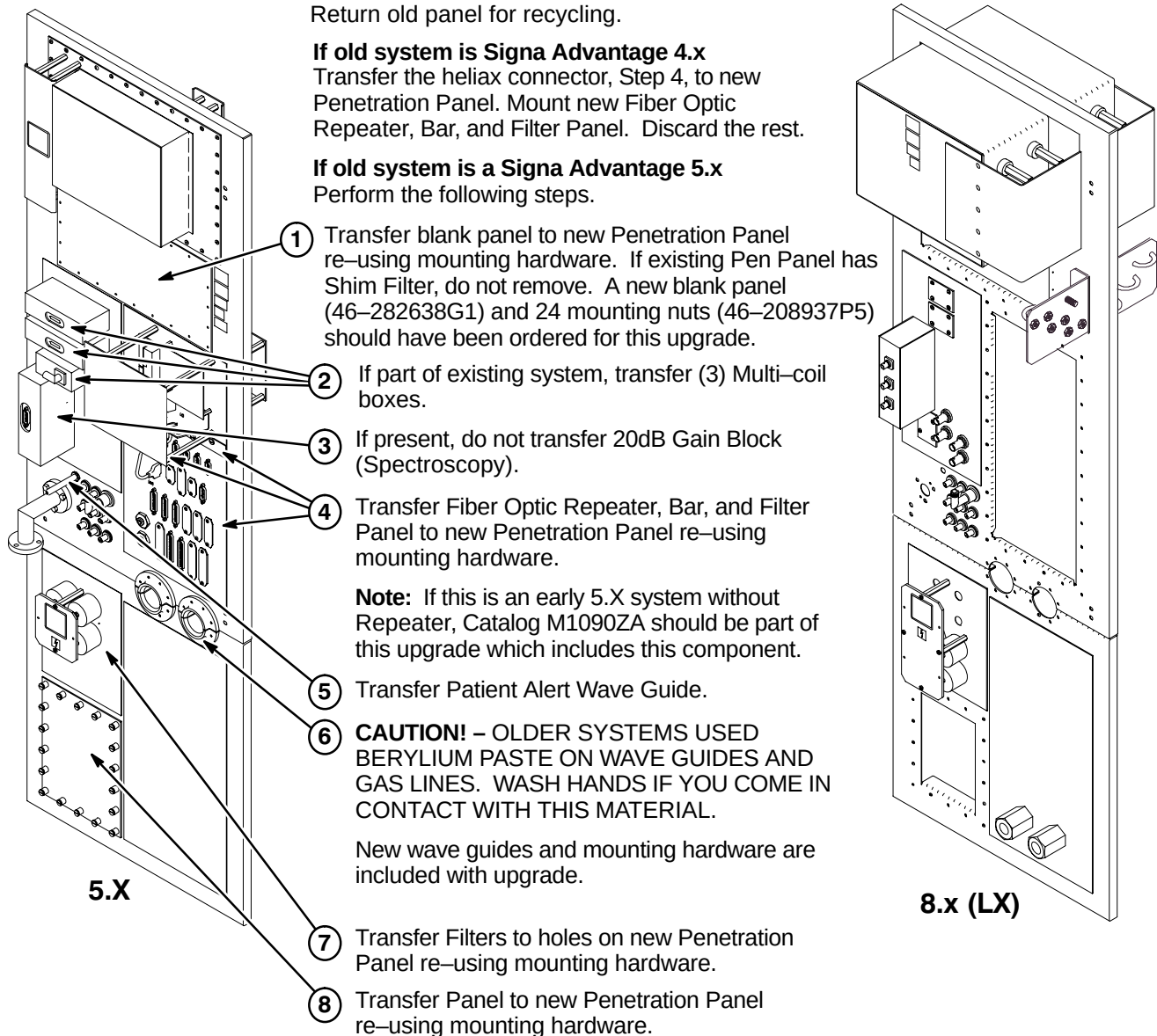
- Customer may have added connections for options to Penetration Panel which are not labelled but must be reconnected on new panel. Deinstall and retain for connection to new Penetration Panel.
- If the site has a Patient Alert Extender Box mounted on or near the Panel, then retain for installation on new Penetration Panel.

- ① If still in place, remove existing Penetration Panel. Install new Penetration Panel using the same mounting hardware of either method shown below.

NOTE: If old system is **Signa 3.x or 3.2 Perf+**
New Penetration Panel should be installed.
Return old panel for recycling.

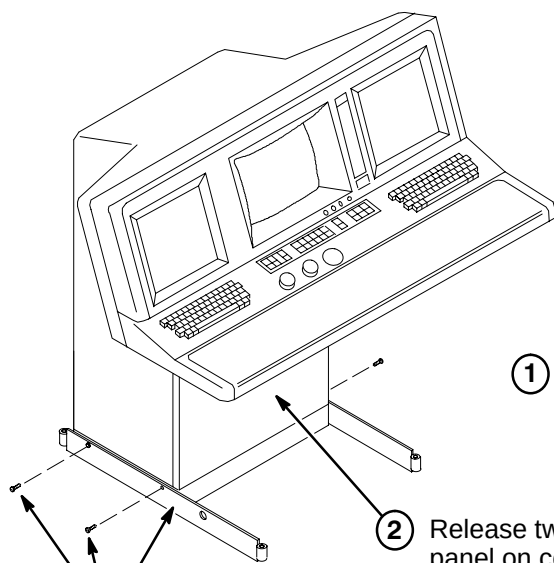
If old system is Signa Advantage 4.x
Transfer the heliax connector, Step 4, to new Penetration Panel. Mount new Fiber Optic Repeater, Bar, and Filter Panel. Discard the rest.

If old system is a Signa Advantage 5.x
Perform the following steps.



5-6 3.x/4.x OPERATOR CONSOLE RETURN PROCEDURE

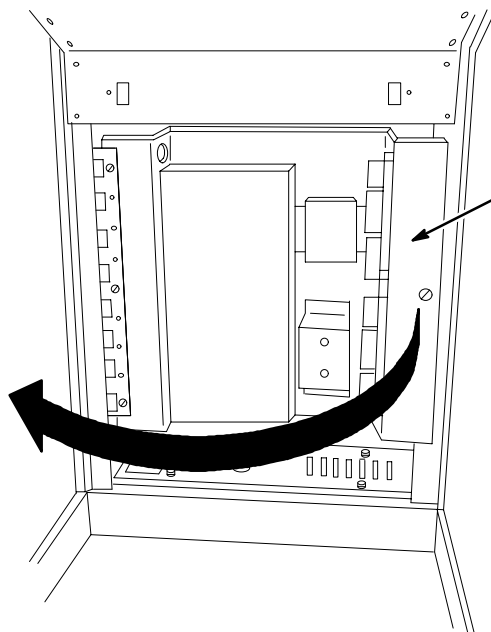
Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of 3.x/4.x Operator Console.

□ A1 – INSTALL OPERATOR CONSOLE SHIPPING LEGS

① Remove any installed Service Keys.

② Release two captive fasteners near top of lower front panel on cosole and remove panel.

③ Fasten to each side of Console:
(1) 46-225881G1 Anti-Tip Shipping Leg
(2) 46-208561P19 1/4-20x 3/4 Long Hex Head Screw

□ A2 – OPERATOR CONSOLE REMOVAL

① Unlatch and swing open panel assembly.

② Remove hardware that attaches console to floor.

③ Close and latch panel assembly.

④ Re-attach Front Panel.

5-7 DEINSTALL 5.x OPERATOR CONSOLE AND RETURN PROCEDURE

Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of 5.x Operator Console.

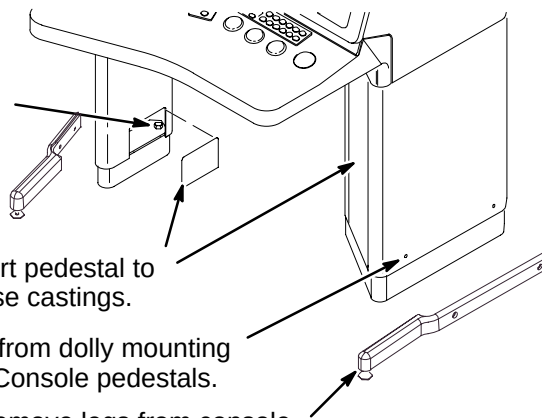
□ A1 – PREPARING OPERATOR CONSOLE REMOVAL

- ④ Loosen screws holding Console to pedestal risers.

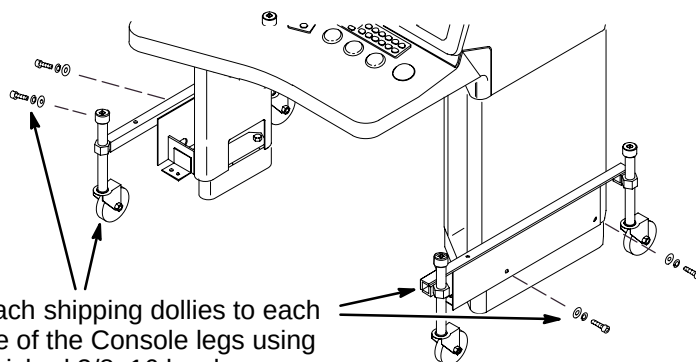
- ③ Remove covers on sides of support pedestal to access mounting holes on the Base castings.

- ② Remove Plug Buttons from dolly mounting holes on each side of Console pedestals.

- ① If installed, unfasten and remove legs from console.

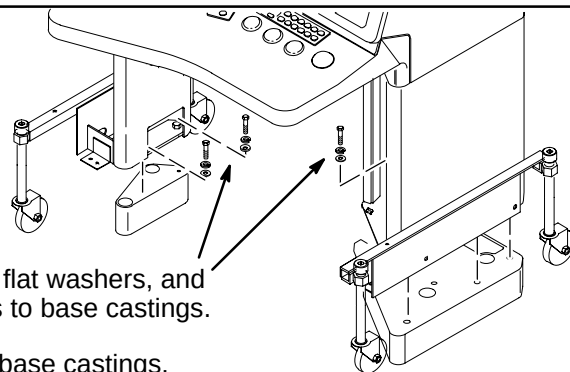
**□ A2 – ATTACHING DOLLY TO OPERATOR CONSOLE**

- ① Attach shipping dollies to each side of the Console legs using furnished 3/8-16 hardware.

**□ A3 – REMOVING OPERATOR CONSOLE**

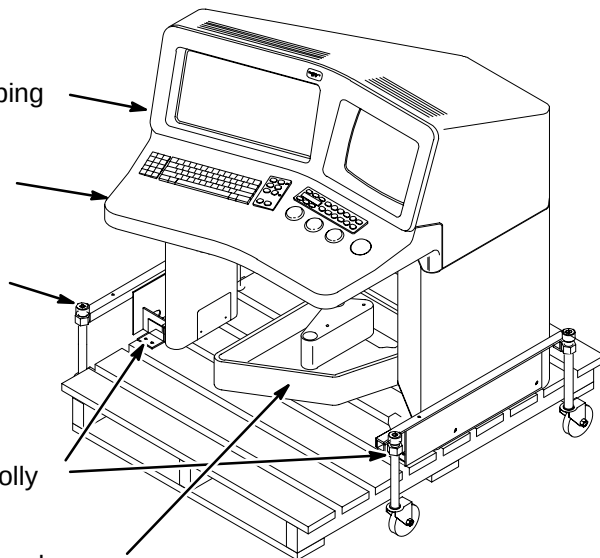
- ① Remove lock washers, flat washers, and screws connecting legs to base castings.

- ② Raise console up over base castings.

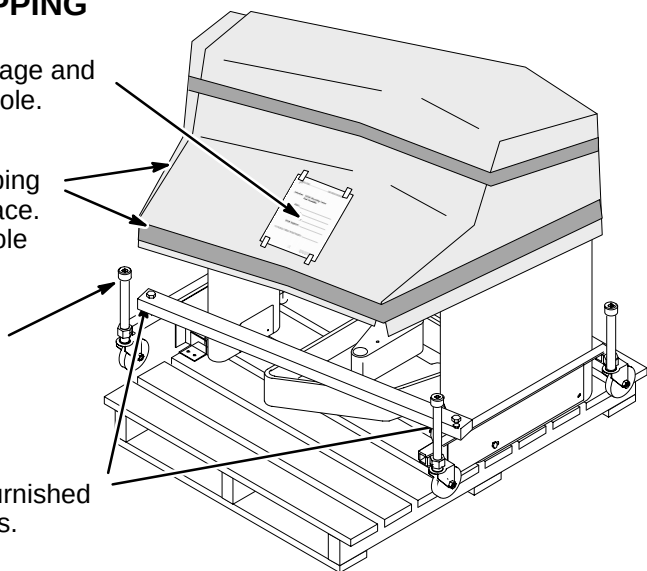


5-7 DEINSTALL 5.x OPERATOR CONSOLE AND RETURN PROCEDURE (Cont'd)**□ A4 – FASTENING OPERATOR CONSOLE TO SHIPPING PALLET**

- ① Carefully roll Operator Console over an available shipping pallet.
- ② Position Console so that Keyboard Housing does not extend beyond shipping pallet.
- ③ Use a 1/2 inch socket wrench to turn the jack screws and lower the Operator Console onto pallet.
- ④ Locate (4) four provided 5/16 x 1-1/2 long lag bolts.
- ⑤ Install two of the lag bolts thru each shipping dolly bracket to fasten to shipping pallet.
- ⑥ Nest together Riser castings on pallet as shown.

**□ A5 – FINAL PREPERATION BEFORE SHIPPING**

- ④ Fill out *GEMS Recycling Form* on next page and attach to top of cardboard covering console.
- ③ Using any available cardboard (i.e. empty shipping boxes), cover Operator Console and tape in place. This is to prevent damage when shipping console back to manufacturer.
- ② Make sure that all four wheel assemblies are completely raised. If not, use 1/2 inch socket wrench to turn jack screws.
- ① Install provided dolly stiffener with furnished 1/4-20 x 1-1/2 long screws and nuts.



Return Console to GE Medical Systems with other system equipment that will be replaced during this upgrade.

FDO: _____

Site Name: _____

Removal Date: _____

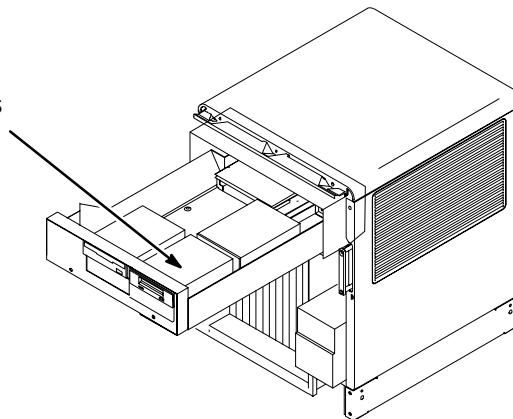
Field Engineer: _____

Genesis Console Serial Number: _____

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5-8 DEINSTALL 5.x COMPUTER AND RETURN PROCEDURE**□ A1 – REMOVAL OF DAT AND MOD DRIVES FROM GENESIS COMPUTER****UPGRADE FROM 5.x OPTIONS:**

Remove Legacy Media DAT and MOD Drives from Genesis Computer for transfer to new Operator Workspace computer.



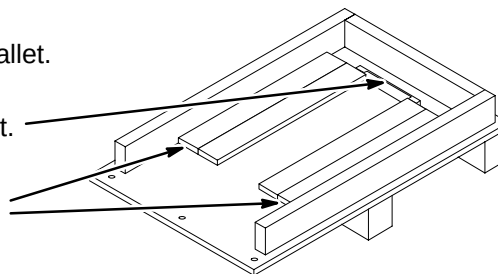
Confirm that all cables are disconnected from Computer. Perform the following procedure for return of 5.x Computer reusing the shipping pallet that the new Operator Workspace Cabinet arrived in.

□ A2 – PREPARING SHIPPING PALLET

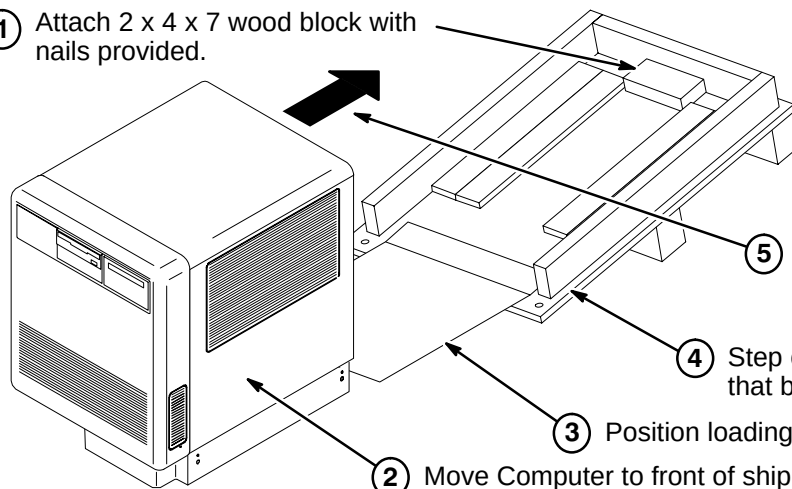
① Locate the Operator Workspace Cabinet shipping pallet.

② Remove existing bracket.

③ Re-position the protective foam on each side from a vertical to a horizontal position between the side of the pallet and the existing horizontal foam.

**□ A3 – LOADING COMPUTER ONTO SHIPPING PALLET**

① Attach 2 x 4 x 7 wood block with nails provided.



Note:
Pallet will return to horizontal position as Computer is rolled toward back of pallet.

⑤ With another person, carefully roll Computer onto shipping pallet so it will rest on the horizontal foam pieces.

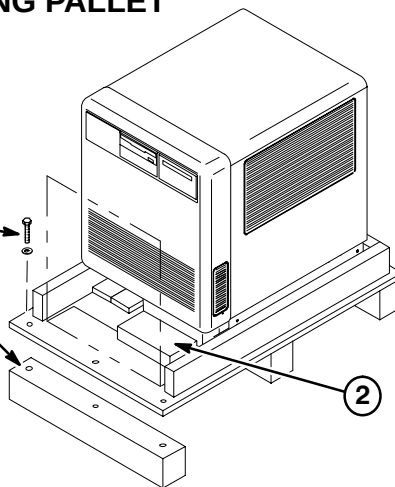
④ Step on front corner of shipping pallet so that bottom board of pallet touches floor.

③ Position loading board at front of pallet.

② Move Computer to front of shipping pallet.

2-3 DEINSTALL 5.x COMPUTER AND RETURN PROCEDURE (Cont'd)**□ A4 – REASSEMBLE SHIPPING PALLET**

- ① Re-attach wood block to pallet using the three previously removed fasteners.

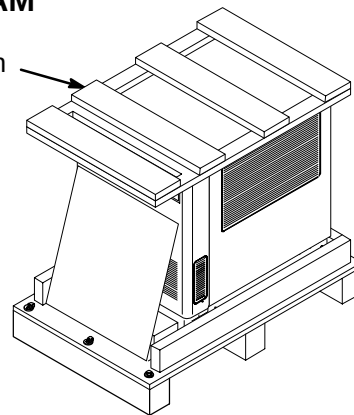


- ② Place 2 x 4 wood block against front right side of Computer and attach to pallet with nails provided.

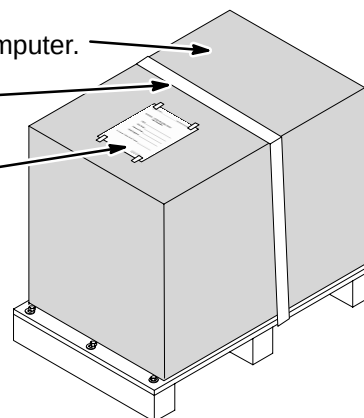
□ A5 – INSTALLING COMPUTER TOP SHIPPING FOAM

- ① Install the Operator Workspace Cabinet shipping foam on top of Computer.

Note: Shipping foam insert will not fit as well as it did with the new Operator Workspace Cabinet.

**□ A6 – FINAL PREPERATION BEFORE SHIPPING**

- ① Place Operator Workspace shipping box over Computer.
- ② Secure to shipping pallet.
- ③ Fill out *GEMS Recycling Form* on next page and attach to top of shipping box.



Return Computer to GE Medical Systems with other system equipment that has been replaced during this upgrade.

FDO: _____

Site Name: _____

Removal Date: _____

Field Engineer: _____

Genesis Computer Serial Number: _____

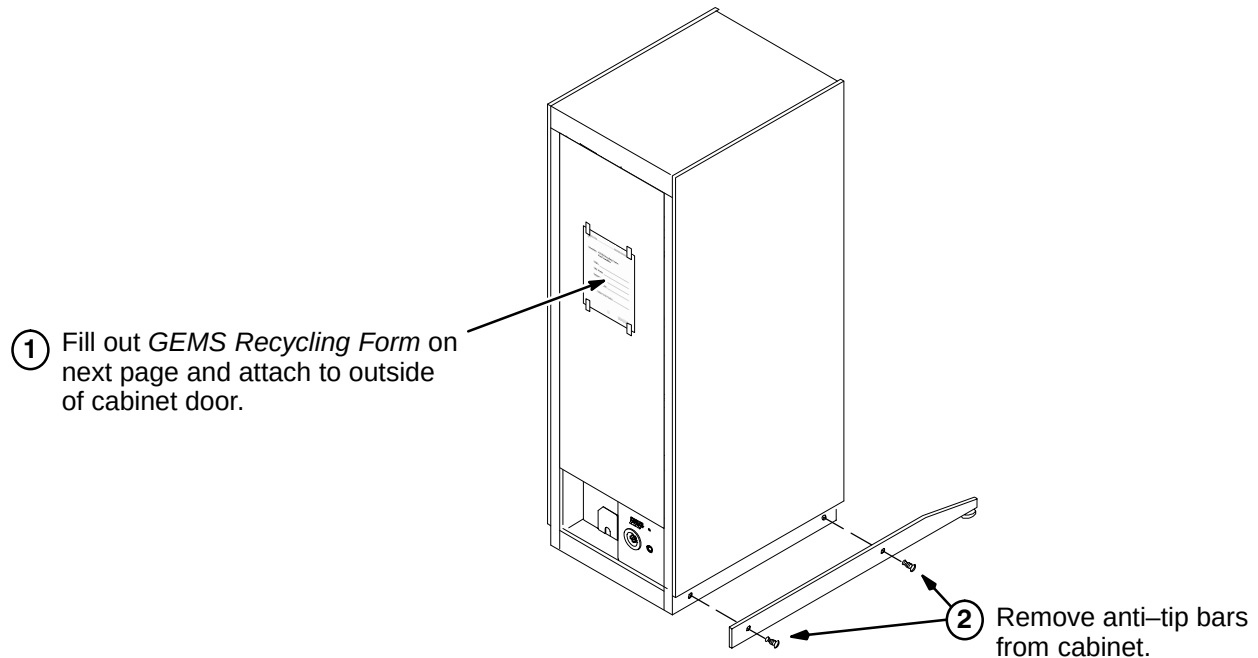
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5-9 CABINET RETURN PROCEDURE

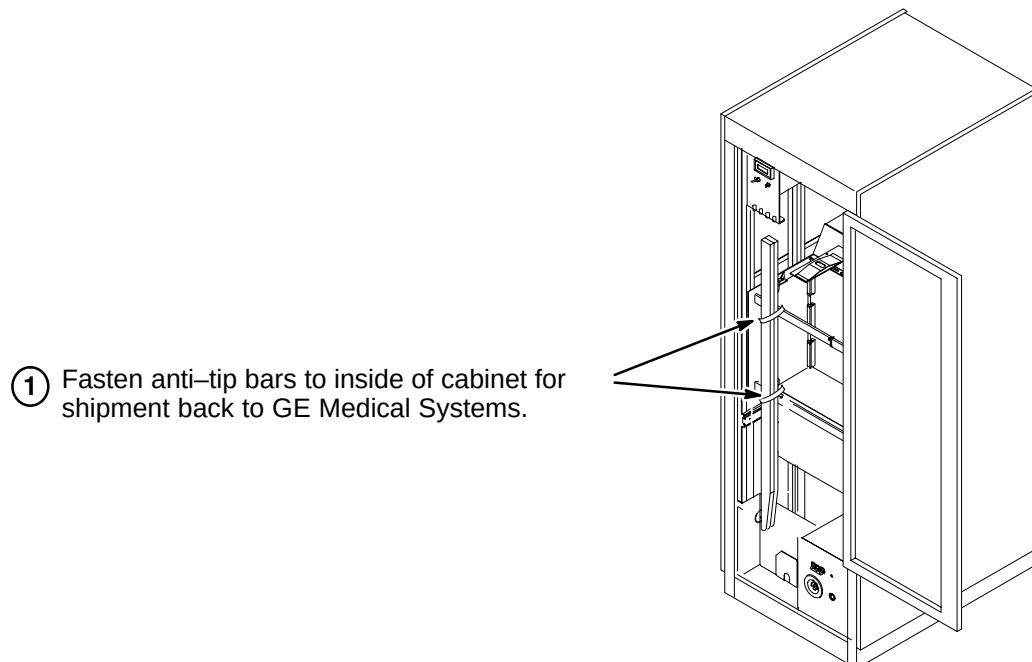
Confirm that all cables are disconnected from applicable Cabinet. The System Cabinet is shown below. The following procedure should be used for return of any cabinet with anti-tip bars.

Refer to following deinstallation for cabinets being removed

□ REMOVING ANTI-TIP BARS



□ ATTACHING ANTI-TIP BARS AND FORM FOR RETURN SHIPMENT



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NOTE: Copy this sheet as needed.

FDO: _____

Site Name: _____

Removal Date: _____

Field Engineer: _____

Cabinet Type: _____

Cabinet Serial Number: _____

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5-10 DEINSTALL SYSTEM CABINET**□ SYSTEM CABINET UPGRADE GUIDELINES**

The Signa Horizon 8.x System Cabinet Upgrade includes a new cabinet which will require items to be transferred from the old cabinet. This Upgrade also provides instructions for other components and options being transferred to the new System Cabinet. Note that various parts will be left over according to the type of system being upgraded. Return to GE these parts.

UPGRADE FROM SIGNA 3.x SYSTEM CABINET

Any Signa 3.x System being upgraded to Signa Horizon 8.x HiSpeed, or EchoSpeed will receive a new System Cabinet with all components installed.

UPGRADE FROM SIGNA ADVANTAGE 4.x SYSTEM CABINET

Any Signa Advantage 4.x System being upgraded to Signa Horizon 8.x HiSpeed or EchoSpeed will receive a new System Cabinet without the TNF Module.

- The new Horizon 8.x System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old two-bay System Cabinet, it will need to be transferred. Refer to procedure on next page for removal. If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade.

UPGRADE FROM SIGNA ADVANTAGE 5.x SYSTEM CABINET

Any Signa Advantage 5.x System being upgraded to Signa Horizon 8.x HiSpeed or EchoSpeed will receive a new System Cabinet without the TNF Module.

- The new Horizon 8.x System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old Advantage 5.x System Cabinet, it will need to be transferred. Refer to procedure on next page for removal. If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade..

UPGRADE FROM SIGNA HORIZON 5.x SYSTEM CABINET

Any Signa Horizon 5.x System being upgraded to Signa Horizon 8.x HiSpeed or EchoSpeed will receive a new System Cabinet without the TNF Module.

- The new Horizon 8.x System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old Horizon 5.x System Cabinet, it will need to be transferred. Refer to procedure on next page for removal. If not present Catalog M1090ZB, M1090ZC, M1090ZF, or M1090ZG should have been ordered prior to upgrade.
- For upgrading from a Horizon 5.x EchoSpeed to a Horizon 8.x Echospeed, the Fast Receiver and cable (2137306) will need to be transferred, refer to procedures on next page for removal.

UPGRADE FROM SIGNA HORIZON 8.x SYSTEM CABINET

Any Signa Horizon 8.x System being upgraded to Signa WideOpen 8.x HiSpeed or EchoSpeed will have Magnet Monitoring installed.

- Remove cryogen meters and related cables. Return with magnet.

5-10 SYSTEM CABINET REMOVAL (continued)**❑ REMOVE TNF MODULE FROM EXISTING CABINET**

For Upgrading from Horizon 5.x: TNF Assembly is present in existing cabinet for transfer. The TNF Expansion Asm (2118107) should also have been installed if Multi-coil is being transferred.

For Upgrading from Advantage 5.x: If present, transfer TNF Assembly Module from existing cabinet. If TNF Assembly Module is not present, order 2118106 for non Multi-coil system or 2118106 and 2118107 for Multi-coil.

For Upgrading from Advantage 4.x: If present, transfer TNF Assembly Module from existing cabinet. If TNF Assembly Module is not present, order 2118106 for non Multi-coil system or 2118106 and 2118107 for Multi-coil.

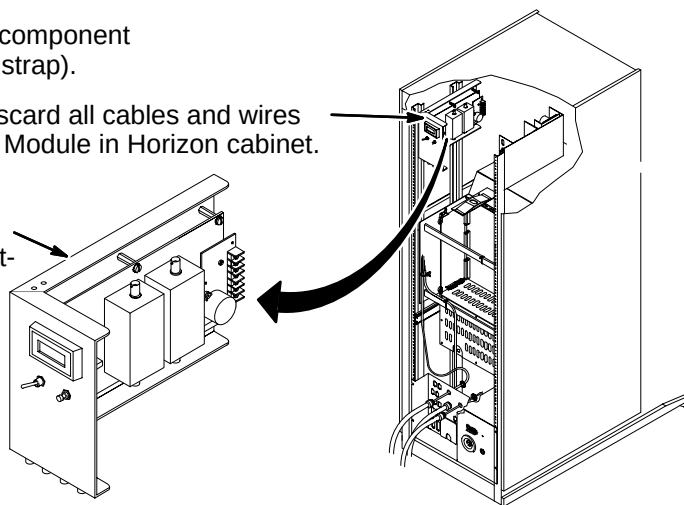
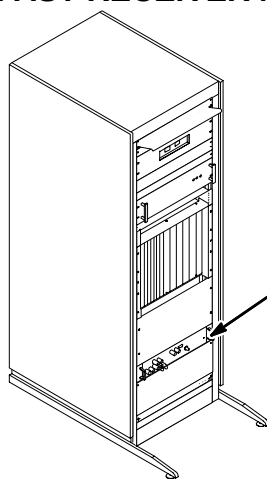
For Upgrading from Advantage 3.x: TNF Assembly should be present in new 8.x cabinet. The TNF Expansion Asm should also have been installed if Multi-coil is part of new 8.x system. If not present, order 2118107.



Note: TNF Module is a static sensitive component and needs to be handled as such (wrist strap).

- ① Disconnect and discard all cables and wires connected to TNF Module in Horizon cabinet.

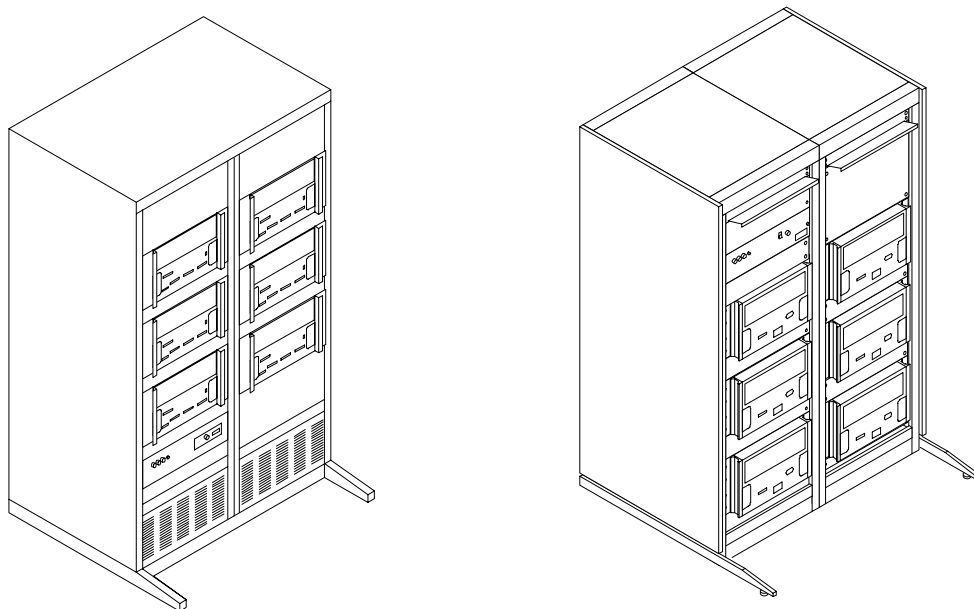
- ② Unfasten and remove TNF Assembly Module (with Expansion Asm if present). Save mounting hardware.

**❑ REMOVE FAST RECEIVER MODULE FROM HORIZON SYSTEM CABINET**

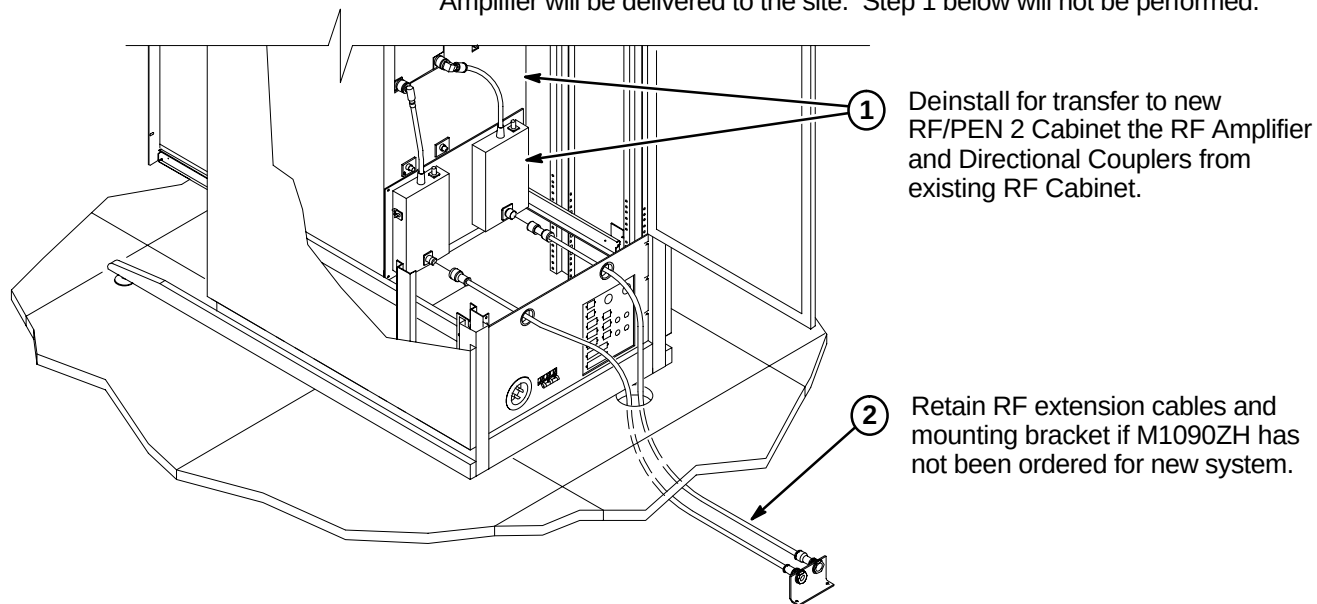
- ① Remove Fast Receiver, rear and front vertical mounting rail brackets, mounting hardware, and slides from Horizon 5.x System Cabinet.

5-11 DEINSTALL PRE-8645 GRADIENT CABINET**□ PRE-8645 GRADIENT CABINET REMOVAL**

- ① Disconnect all cables. Cabinet to be returned.

**5-12 DEINSTALL RF CABINET****□ D – RF CABINET CABLE CONNECTIONS**

Note: For a 3.x to 8.x WideOpen upgrade, a new RF/Pen Cabinet with RF Amplifier will be delivered to the site. Step 1 below will not be performed.



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5-13 DEINSTALLATION FEEDBACK DATA SHEET

Use this form to list the problems you had with the Deinstallation at this site.

Site Name: _____

Location: _____

Magnet Type: _____ (S2, S3, S4 etc...)

System Type: _____ (3x, 4x, 5x, other)

Problems encountered with desinstall procedures and Flowcharts

Site related problems

List any cables that were cut or damaged during deinstall. List any hardware that sustained any major damage during deinstall:

Send this completed form to:

Magnet Deinstall Feedback
MR Service Engineering, W827
GE Medical Systems
P.O. Box 414
Milwaukee, Wisconsin 53201-9824